arXiv:2506.22343v1 [stat.ML] 27 Jun 2025

Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts

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June 27, 2025

Abstract

Text watermarks in large language models (LLMs) are an increasingly important tool for detecting synthetic text and distinguishing human-written content from LLM-generated text. While most existing studies focus on determining whether entire texts are watermarked, many real-world scenarios involve mixed-source texts, which blend human-written and watermarked content. In this paper, we address the problem of optimally estimating the watermark proportion in mixed-source texts. We cast this problem as estimating the proportion parameter in a mixture model based on *pivotal statistics*. First, we show that this parameter is not even identifiable in certain watermarking schemes, let alone consistently estimable. In stark contrast, for watermarking methods that employ continuous pivotal statistics for detection, we demonstrate that the proportion parameter is identifiable under mild conditions. We propose efficient estimators for this class of methods, which include several popular unbiased watermarks as examples, and derive minimax lower bounds for any measurable estimator based on pivotal statistics, showing that our estimators achieve these lower bounds. Through evaluations on both synthetic data and mixed-source text generated by open-source models, we demonstrate that our proposed estimators consistently achieve high estimation accuracy.

1 Introduction

Over the past two years, the rapid advancement of large language models (LLMs) has profoundly transformed content generation, with diverse applications spanning creative writing, code generation, text summarization, and educational assistance [23]. Their rapid adoption, however, has raised pressing concerns regarding content authenticity in critical domains such as scientific peer review [21], education [22], journalism [29], and data authenticity [26, 7].

To mitigate potential misuse of these powerful generative tools, researchers have developed *text watermarking* methods that embed imperceptible yet statistically verifiable signals into LLM-generated output [15, 1, 17]. These watermarking techniques facilitate reliable detection of synthetic text, enabling one to distinguish fully watermarked content from human-written text [24].

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However, real-world scenarios frequently involve mixed-source texts, where human-written and LLM-generated content are commonly interwoven [40, 12, 19]. This is particularly evident in academic peer reviews, as demonstrated by Liang et al. [21], who reported a growing prevalence of LLM-modified reviews in machine learning conferences. Beyond academia, evidence suggests that human-AI collaboration is increasingly reshaping content creation practices across various domains [3]. In this context, quantifying the extent of human versus LLM contribution becomes crucial [39]. Indeed, accurate estimates of human-written proportions have significant implications for determining content ownership and authenticity in academic writing, education, and legal documents [37]. For instance, a homework assignment essay with only 10% student contribution might be deemed unacceptable, while one with 80% or more student input might be considered permissible.

While prior studies have primarily focused on binary detection of fully watermarked versus fully human-written texts [20], recent work has begun to address the challenges posed by mixed-source content. From a statistical standpoint, however, the problem of estimating the proportion of LLM-generated content in mixed-source texts remains insufficiently explored, despite its critical role in use cases such as authorship attribution. Notably, Li et al. [19] formulated watermark detection in mixed-source texts as a mixture detection problem and identified optimal detection methods through the lens of statistical robustness. Other studies have examined the more granular problem of identifying watermarked segments within mixed-source texts. Examples include Aaronson [1], Kirchenbauer et al. [16]'s continuous span search, Zhao et al. [44]'s online learning approach, and Li et al. [18]'s application of change point detection to identify watermarked segments.

While identifying watermarked segments can, in principle, suffice for our purpose, it often yields an unnecessarily fine level of granularity and, more critically, may be statistically infeasible for pinpointing precisely which segments are human-authored. In many scenarios, a single numerical estimate of the proportion of LLM-generated content is more convenient and interpretable. For instance, it is practical to require AI-generated content to not exceed a 10% threshold in critical documents such as legal briefs and academic papers, while a higher limit (e.g., 50%) may be tolerable in less critical contexts such as homework assignments. Moreover, precisely localizing human-written text is arguably more challenging than simply estimating the overall proportion. This parallels findings from the broader mixture detection literature, wherein identifying individual signal instances is inherently more difficult than detecting their aggregate presence. Indeed, signals can be statistically detectable even when it is information-theoretically impossible to locate each instance accurately [10, 5].

1.1 Our Contributions

In this paper, we address the problem of estimating the proportion of tokens—smallest units in LLM generation that, without loss of generality, can be thought of as words—generated by watermarked LLMs in mixed-source texts under general watermarking schemes. Our main contributions are:

Problem formulation and identifiability: We formulate the estimation problem as a mixture model parameter estimation task using pivotal statistics [20]. Roughly speaking, a pivotal statistic quantifies the evidence that a given word (or token) is watermarked. Through rigorous analysis, we show that the proportion parameter in green-red list watermarking schemes [15] is not identifiable, which leads to the inherent bias in traditional approaches like maximum likelihood estimation (MLE). However, for watermarks with continuous pivotal statistics, we establish conditions under which identifiability can be achieved.

New estimators with theoretical guarantees: We develop computationally efficient estimators

for identifiable cases and provide theoretical guarantees for their estimation accuracy. Our comprehensive experimental validation, conducted on both synthetic data and mixed-source texts from open-source LLMs, shows the consistent and superior performance of our proposed estimators.
Matched minimax lower bounds: We establish minimax lower bounds for broad classes of measurable estimators and prove that our proposed methods generally achieve these bounds, thereby establishing their theoretical optimality.

1.2 Related Work

Most watermarking studies focus on the binary detection problem, which aims to determine whether a text is fully watermarked or entirely human-written. However, this narrow scope fails to address real-world scenarios where texts often originate from mixed sources. While recent work has begun exploring the detection of watermarked segments in mixed-source texts, their approaches differ significantly from ours. Li et al. [19] investigated the optimal detection of watermarked segments under human editing, assuming that the text is initially fully generated by a watermarked LLM and later modified by a human editor. Using the Gumbel-max watermark as an example, they demonstrate that a family of truncated goodness-of-fit tests achieves optimal detection, where optimality is defined by the phase transition boundary and feast-favorable efficiency notion introduced in [20]. However, their work focuses exclusively on detection and does not address estimation.

Other researchers attempt to locate watermarked segments by assigning scores to individual tokens or intervals of tokens and identifying watermarked (or LLM-generated) content based on extreme values. For example, Kirchenbauer et al. [16] searched for the largest intervals or continuous token spans with the highest z-scores by testing all possible window sizes and traversing the entire text for each size, resulting in a computational complexity of $\widetilde{O}(n^2)$. Aaronson [1] improved on this by using dynamic programming to identify the largest continuous token span, achieving a reduced complexity of $O(n^{1.5})$. Zhao et al. [44] defined token scores based on the expectations of pivotal statistics and employs an online learning algorithm for estimation, achieving a complexity of $O(n \log n)$ for texts of length n. Inspired by change point detection, Li et al. [18] used bootstrap-based p-values as token scores, but the method incurs a higher computational cost of approximately $\widetilde{O}(n^2)$ due to its analysis of dependencies across all sliding token windows and pseudorandomness.

Despite these advancements, existing methods neither address the problem of watermark proportion estimation nor provide a systematic analysis of their performance or hyperparameter tuning (e.g., threshold selection). In contrast, our work presents the first comprehensive framework for watermark proportion estimation. We address key questions such as what to estimate (estimand and estimability), how to estimate (estimation methods), how accurately it can be estimated (estimation accuracy), and the theoretical limits of estimation (minimax lower bounds). Furthermore, our proposed estimator achieves $O(n)$ complexity by requiring only a logarithmic number of passes through the samples.

2 Preliminaries

LLMs, such as the GPT series [25, 4], generate text by autoregressively sampling discrete units called tokens. Let $\mathcal{W}$ denote the vocabulary (token universe), and let $w \in \mathcal{W}$ be a single token. At each step t, given the prefix $w_{1:(t-1)} := w_1 w_2 \cdots w_{t-1}$, the model computes a multinomial distribution $\boldsymbol{P}_t := (P_{t,w})_{w \in \mathcal{W}}$ over the vocabulary, and samples the next token $w_t \sim \boldsymbol{P}_t$. This distribution $\boldsymbol{P}_t$,

called the next-token prediction (NTP) distribution, is unknown to the verifier, as it depends on the user-supplied prompt, and potentially hidden system prompts [32].

A general framework for watermarking. We adopt a general three-party framework that reflects real-world applications of watermark-based verification [17, 38]. The language model provider (e.g., OpenAI) embeds statistical watermark signals into the generated text using a designated watermarking scheme. Under this setup, every token generated by the model is watermarked by design. The user (e.g., a student) interacts with the model, receives the watermarked output, and may optionally modify the text before sharing or submitting it. The verifier (e.g., a teacher) is tasked with estimating how many tokens in a given document originated from the watermarked LLM.

Watermarking schemes operate by modifying the standard sampling process to embed statistical signals that can later be recovered for verification. This is typically achieved by introducing a pseudorandom variable ζ_t at each step t, which serves as a random seed that guides token selection. Instead of sampling w_t directly from $\boldsymbol{P}_t$, the model applies a decoding rule $w_t := \mathcal{S}(\boldsymbol{P}_t, \zeta_t)$, where $\mathcal{S}$ is a (possibly stochastic) decoding function. The pseudorandom variable ζ_t is generated using cryptographic techniques and can be deterministically reconstructed by the verifier if the construction is known.[1] In this work, we assume ζ_t are i.i.d. samples from a known distribution, consistent with the perfect pseudorandomness assumption used in prior work [36, 43]. A watermark is called unbiased if the resulting token still marginally follows the original NTP distribution $\boldsymbol{P}_t$; otherwise, it is biased [20].

In contrast, human-written tokens are also drawn from $\boldsymbol{P}_t$ but do not depend on ζ_t, leading to a different joint distribution over (w_t, ζ_t). Detection thus reduces to testing whether w_t and ζ_t are statistically dependent. For human-written text, they are independent, since human users know nothing about the pseudorandomness; for watermarked text, dependence is introduced through the decoding rule $\mathcal{S}$. To detect this dependence, Li et al. [20] proposed computing a pivotal statistic $Y_t = Y(w_t, \zeta_t)$, designed to control Type I error and boost power. These statistics are scalars that summarize the degree of dependence between w_t and ζ_t, which follow a known distribution under the null (i.e., no watermark). Specifically, $Y_t \sim \mu_0$ i.i.d. when no watermark is present, and $Y_t \mid \boldsymbol{P}_t \sim \mu_{1,\boldsymbol{P}_t}$ under the alternative due to watermark signals. Many watermarking schemes can be cast in this form, and detection is based solely on the sequence $Y_{1:n}$. This approach enables the verifier to statistically infer the presence and extent of watermarked content, even without access to the original prompt or NTP distributions.

Three representative watermarks. We study three representative watermarking schemes under the general decoding framework described above. These schemes differ primarily in the choice of decoder $\mathcal{S}$ and the construction of the pseudorandom number ζ_t. Once these components are specified, the watermarking scheme is fully determined. For detection purposes, once the pivotal statistics are defined, the null distribution μ_0 and the alternative distributions $\mu_{1,\boldsymbol{P}}$ are also fixed. Therefore, it suffices to highlight these structural differences. Table 1 summarizes the key components of each scheme.

We briefly introduce each scheme below. The Gumbel-max watermark [1], recognized as the first unbiased watermark, has been implemented internally at OpenAI. It uses the Gumbel-max trick [13] to sample tokens, which by construction ensures unbiasedness. The inverse transform watermark [17]

[1] For example, $\zeta_t = \mathcal{A}(w_{(t-m):(t-1)}, \texttt{Key})$ for some hash function $\mathcal{A}$, context window size m, and secret key `Key`. Given the full text $w_{1:n}$, the hash function, and the key, one can reconstruct $\zeta_{1:n}$.

Table 1: Summary of three watermarks.

Watermark	**Pseudorandom** ζ	**Decoder** $\mathcal{S}(\boldsymbol{P}, \zeta)$	**Pivotal statistic** $Y(w, \zeta)$	μ_0	$\mu_{1,\boldsymbol{P}}$
Gumbel-max [1]	$\zeta = (U_1, \ldots, U_{\lvert\mathcal{W}\rvert})$ with each U_w i.i.d. from $U(0,1)$	Gumbel-max trick: $\arg\max_w \frac{\log \zeta_w}{P_w}$ (which follows $\boldsymbol{P}$)	U_w	$U(0,1)$	The CDF is: $\sum_w P_w r^{1/P_w}$ with $r \in [0,1]$ [20]
Inverse transform [17]	$\zeta = (U, \pi)$ with $U \sim U(0,1)$ and π a permutation on $\mathcal{W}$	$\pi^{-1} \circ F_\pi^{-1}(U)$ where F_π^{-1} is the generalized inverse of F_π	$1 - \lvert U - \eta_\pi(w)\rvert$ where $\eta_\pi(w) = \frac{\pi(w)-1}{\lvert\mathcal{W}\rvert-1}$	Asymp. CDF: r^2 with $r \in [0,1]$	Asymp. CDF is $(1 - \frac{1-r}{P_{(1)}})^2$ with $r \in [1 - P_{(1)}, 1]$ [20]
Green-red list [15]	$\zeta = \mathcal{G}$ which is a random set of $\mathcal{W}$ with $\lvert\mathcal{G}\rvert = \lvert\mathcal{W}\rvert \cdot \gamma$	sampled from $\boldsymbol{P}_{gr}$ where $(P_{gr})_w \propto P_w \cdot \exp(\delta \mathbf{1}_{w \in \mathcal{G}})$	$\mathbf{1}_{w \in \mathcal{G}}$, whether the token w is in the green set $\mathcal{G}$.	$\mathrm{Ber}(\gamma)$	$\mathrm{Ber}(\mu)$ where $\mu = \frac{\sum_{w \in \mathcal{G}} P_w \exp(\delta)}{\sum_{w \in \mathcal{G}} P_w \exp(\delta) + \sum_{w \notin \mathcal{G}} P_w}$ [15]

applies inverse transform sampling to the CDF $F_\pi(x) = \sum_w P_w \mathbf{1}_{\pi(w) \le x}$, which defines a distribution over the vocabulary $\mathcal{W}$ based on a permutation π. The green-red list watermark, though biased, is widely used due to its simplicity. It partitions the vocabulary into green and red token sets, secretly boosts the probability of green tokens, and signals watermarking if the observed frequency of green tokens exceeds an expected threshold. Follow-up work has improved this scheme through parameter tuning [35] and by developing unbiased [14, 36, 38] or robust variants [42, 45].

3 Proportion Estimation Methods

In this section, we study how to estimate the proportion of watermarked content in mixed-source text. Section 3.1 defines the problem, Section 3.2 analyzes its estimability, and Section 3.3 presents our estimators.

3.1 Target Estimand

The first step is to identify the quantity we aim to estimate and the specific model where it arises. Most existing watermark detection methods rely on pivotal statistics [20], which provide quantitative evidence of watermark signals at each token–pseudorandom pair (w_t, ζ_t). Those statistics exhibit a key dichotomy based on watermark integrity. When the watermark signal in (w_t, ζ_t) is disrupted—such as by modifying w_t or altering its preceding tokens so that ζ_t no longer generates w_t—w_t becomes statistically independent of ζ_t due to the statistical sensitivity of hash functions. In such cases, the pivotal function Y ensures that $Y_t = Y(w_t, \zeta_t)$ always follows a known distribution μ_0, regardless of the distribution of w_t. Conversely, if the watermark signal in (w_t, ζ_t) remains intact, w_t continues to depend on ζ_t via the decoder $\mathcal{S}$ so that $Y_t \sim \mu_{1,\boldsymbol{P}_t}$ with $\boldsymbol{P}_t$ the NTP distribution for w_t. So, given $\boldsymbol{P}_t$, Y_t follows either the null μ_0 or the alternative $\mu_{1,\boldsymbol{P}_t}$. The expected proportion following the alternative distributions naturally becomes our target estimand, as in Model 1.

Model 1 (Li et al. [19]). *The pivotal statistic Y_t satisfies*

$$Y_t | \boldsymbol{P}_t \sim (1 - \varepsilon)\mu_0 + \varepsilon \mu_{1,\boldsymbol{P}_t}, \quad \forall t \in [n], \tag{1}$$

where ε represents the proportion of watermark signals, our estimand of interest. Rigorously, there exists a σ-field $\{\mathcal{F}_t\}_{t \ge 1}$ so that $\boldsymbol{P}_t \in \mathcal{F}_{t-1}$ and $Y_t \in \mathcal{F}_t$ with

$$\mathbb{E}[\mathbf{1}\{Y_t \le x\} | \mathcal{F}_{t-1}] = (1 - \varepsilon)F_0(x) + \varepsilon F_{\boldsymbol{P}_t}(x).$$

Here, $\boldsymbol{P}_t$ depends on the history of tokens $w_{1:(t-1)}$, so we view it as $\mathcal{F}_{t-1}$-measurable.

Remark 3.1. *The σ-field $\mathcal{F}_t$ represents the full information available up to token t, including the generation history $w_{1:t}$ and the NTP distributions $\boldsymbol{P}_{1:(t+1)}$. The formulation in (1) implies that, conditioned on $\boldsymbol{P}_t$, the pivotal statistic Y_t follows a mixture of the null distribution μ_0 and the alternative $\mu_{1,\boldsymbol{P}_t}$. While $\mathcal{F}_t$ is a useful concept, it includes unobservable components like $\boldsymbol{P}_t$, which the verifier does not have access to in practice. This model thus reflects the verifier's uncertainty over user edits without assuming any specific editing strategy.*

While Model 1 accurately captures the autoregressive behavior of LLMs, its application in practice and theory is constrained by two factors. First, from a practical standpoint, the verifier does not have access to critical components of the filtration $\mathcal{F}_t$, particularly each NTP distribution $\boldsymbol{P}_t$, which depend on the original prompt and the internal state of the LLM. This makes it infeasible to use the exact filtration in any detection or estimation procedure. Second, from a theoretical standpoint, the fact that each $\boldsymbol{P}_t$ evolves with the token history introduces complex temporal dependencies. Analyzing this dependence typically requires advanced martingale-based techniques, as seen in prior work [20, 19]. These challenges motivate the use of a simplified surrogate model, which we describe next.

Model 2 (Surrogate model)**.** *We assume that each $\boldsymbol{P}_t$ is fixed at the start of a generation, and the pivotal statistic Y_t is still generated according to* (1)*. In this setup, $Y_1, \ldots, Y_n$ are independent but not identically distributed.*

The key difference between Model 2 and Model 1 is the removal of the statistical dependence of the NTP distributions on token history. We argue that Model 2 is a realistic and reasonable simplification—primarily because, from the verifier's perspective, Models 1 and 2 are operationally indistinguishable. This indistinguishability arises from the fact that the verifier lacks access to critical internal information used during generation, such as the original prompt, the internal state of the language model, and the NTP distributions themselves. This can be further illustrated through the following thought experiment. Consider a sequence $w_{1:n}$ generated under Model 1, where each $\boldsymbol{P}_t$ depends on the full token history: $w_t = \mathcal{S}(\boldsymbol{P}_t, \zeta_t)$ with $\boldsymbol{P}_t = \mathrm{LLM}(w_{1:(t-1)})$. Now imagine freezing this exact sequence $\boldsymbol{P}_{1:n}$ and defining a new model, $\mathrm{LLM}_{\mathrm{new}}$, that always uses the pre-computed $\boldsymbol{P}_t$ at step t to sample tokens regardless of the generation history. If $w_{1:n}$ is produced by LLM, we are in Model 1; if by $\mathrm{LLM}_{\mathrm{new}}$, we are in Model 2. Although the internal mechanisms differ, both models yield the same tokens and pivotal statistics—making them indistinguishable to the verifier, who observes only the final tokens and pseudorandom numbers, but not the model internals. For this reason, we adopt Model 2 for method development and theoretical analysis, while evaluating our methods under the more realistic Model 1 in experiments. As we will show, the estimators derived under Model 2 remain robust and effective when applied to data generated under Model 1.

Remark 3.2. *It is worth noting that the watermark proportion ε is defined in terms of pivotal statistics rather than individual tokens. As introduced in Section 2, each pivotal statistic is a deterministic function of consecutive tokens. As a result, modifying a single token may affect multiple pivotal statistics, leading to a discrepancy between the proportion of watermarked tokens and that of watermarked pivotal statistics. We adopt this definition for its simplicity and because nearly all existing detection rules rely on pivotal statistics.*

3.2 Estimability

The next question is whether the proportion ε is well-defined—or, more precisely, identifiable—when the NTP distributions $\boldsymbol{P}_{1:n}$ are unknown and potentially vary, as is the case for real-world verifiers. Identifiability, in this context, means that ε can be uniquely determined from the observed pivotal statistics alone, even when $\boldsymbol{P}_{1:n}$ are unknown and arbitrary. If ε is not identifiable, multiple values of ε could yield the same data distribution, making it impossible to obtain a unique and accurate estimate. This ambiguity challenges the interpretability of any downstream use of the estimate, as its real-world meaning is unclear.

Impossibility for the green-red list watermark. Unfortunately, for the green-red list watermark, the proportion ε is not identifiable. To illustrate this, consider a straightforward example of proportion estimation in an i.i.d. binary mixture model.

Lemma 3.1. *If data points are i.i.d. from the binary mixture* $(1-\varepsilon)\mathrm{Ber}(\gamma)+\varepsilon\mathrm{Ber}(\mu)$ *where both* ε *and* μ *are unknown while* γ *is known, then* ε *is not identifiable.*

The real-world use of green-red list watermarks is more complex than the simplified setting in Lemma 3.1. In practice, each $Y_t \mid \boldsymbol{P}_t \sim (1-\varepsilon)\mathrm{Ber}(\gamma)+\varepsilon\mathrm{Ber}(\mu_t)$, where μ_t depends on the unknown NTP distribution $\boldsymbol{P}_t$. The expression for μ_t is given in Table 1. As a result, we arrive at the following corollary. The non-identifiability in this case arises because each Y_t is binary and thus carries limited information about the underlying parameters.

Corollary 1. *Under Model 2,* ε *is non-identifiable for the green-red list watermark.*

Identifiability for continuous pivotal statistics. Interestingly, the estimation problem becomes fundamentally different when Y_t is continuous. Specifically, by assuming $\mu_{1,\boldsymbol{P}}$ differs uniformly from μ_0 as in Assumption 3.1, ε becomes identifiable (Lemma 3.2).

Assumption 3.1. *Let* $F_0(x)$ *and* $F_{\boldsymbol{P}}(x)$ *denote the CDF of* μ_0 *and* $\mu_{1,\boldsymbol{P}}$*, respectively, i.e.,* $F_0(x) = \mu_0(Y \le x)$ *and* $F_{\boldsymbol{P}}(x) = \mu_{1,\boldsymbol{P}}(Y \le x)$*. We assume that (i)* F_0 *and* $F_{\boldsymbol{P}}$ *are differentiable everywhere, (ii)* $\lim_{x\to 0} F_{\boldsymbol{P}}(x)/F_0(x) = 0$ *for any non-singular* $\boldsymbol{P}$ *(i.e.,* 1 *is not an entry of* $\boldsymbol{P}$*), and (iii) all of NTP distributions* $\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n$ *are non-singular.*

Lemma 3.2. *Under Assumption 3.1 and Model 2,* ε *is identifiable.*

This result applies to both the Gumbel-max and inverse transform watermark schemes, as their pivotal statistics are continuous and satisfy Assumption 3.1. As a result, ε is identifiable for these two watermarks.

Lemma 3.3. *Assumption 3.1 holds for the Gumbel-max and inverse transform watermark.*

Remark 3.3. *Due to the intricate permutation, the inverse transform watermark satisfies Assumption 3.1 asymptotically, where* $P_{(2)} \cdot \log|\mathcal{W}| \to 0$ *as* $|\mathcal{W}| \to \infty$ *[19]. This condition holds in our simulation study, while it may not be satisfied in practice.*

3.3 Estimation Methods

The proportion ε satisfies the equation

$$\bar{F}(x) = (1-\varepsilon)F_0(x) + \varepsilon \bar{F}_{\boldsymbol{P}}(x) \text{ for any } x, \tag{2}$$

where $\bar{F}(x) = \frac{1}{n}\sum_{t=1}^{n} \mathbb{P}(Y_t \le x)$ is the averaged CDF of observed Y, $F_0(x)$ is the null CDF, and $\bar{F}_{\boldsymbol{P}}(x) = \frac{1}{n}\sum_{t=1}^{n} F_{\boldsymbol{P}_t}(x)$ is the averaged alternative CDF. As a reminder, under Model 2, $\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n$ are fixed, unknown, and predetermined so that $\bar{F}_{\boldsymbol{P}}$ is free of randomness. Equation (2) implies that for any weight function $v : \mathbb{R} \mapsto \mathbb{R}$, it must satify

$$\mathbb{E}_{\bar{F}}[v] = (1-\varepsilon)\,\mathbb{E}_{F_0}[v] + \varepsilon\,\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v],$$

where expectations $\mathbb{E}_{\bar{F}}[v]$, $\mathbb{E}_{F_0}[v]$, and $\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]$ represent integrals with respect to $\bar{F}$, F_0, and $\bar{F}_{\boldsymbol{P}}$, respectively. Rearranging this expression gives

$$\varepsilon = \frac{\mathbb{E}_{F_0}[v] - \mathbb{E}_{\bar{F}}[v]}{\mathbb{E}_{F_0}[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]}. \tag{3}$$

Equation 3 forms the basis for estimators inspired by moment methods [2], which match observed sample moments with theoretical ones. Here, $\mathbb{E}_{F_0}[v]$ is computable because F_0 is known or can be approximated through Monte Carlo simulation. For the observed mixture samples $Y_1, \ldots, Y_n$, $\mathbb{E}_{\bar{F}}[v]$ is estimated by the sample average $\frac{1}{n}\sum_{t=1}^{n} v(Y_t)$. However, the term $\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]$ is more challenging because $\bar{F}_{\boldsymbol{P}}$ is unknown and depends on the NTP distributions $\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n$, which vary across tokens. Estimating these NTP distributions directly is impractical due to their high dimensionality and variability. For example, in models like OPT-1.3B, the vocabulary size $|\mathcal{W}| = 50,272$ [41], meaning each $\boldsymbol{P}_t$ is a distribution over tens of thousands of tokens. With text lengths n often around 100, the number of parameters far exceeds the number of samples, making estimation infeasible.

Initial attempt by ignoring $\bar{F}_{\boldsymbol{P}}$. To simplify, we initially ignore $\bar{F}_{\boldsymbol{P}}$ by choosing the weight function $v_{\text{ind}}(x) = \mathbf{1}\{x \le \delta\}$, where $\delta > 0$ is a small constant specified by the verifier. For small δ, the unknown term $\bar{F}_{\boldsymbol{P}}(\delta)$ becomes negligible compared to $F_0(\delta)$ because Assumption 3.1 requires $\lim_{\delta \to 0} \frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)} = 0$. This leads to the following estimator:

$$\widehat{\varepsilon}_{\text{ini}}(\delta) = 1 - \frac{\widehat{F}(\delta)}{F_0(\delta)}, \tag{4}$$

where $\widehat{F}(x) = \frac{1}{n}\sum_{t=1}^{n} \mathbf{1}\{Y_t \le x\}$ is the empirical CDF of the observed pivotal statistics. While this estimator is simple, it has two significant limitations. First, relying on a small δ means only a few samples are used which reduces statistical efficiency. For example, if δ is chosen too small, most observed values Y_t may exceed δ, leading to unstable estimates due to limited sample sizes. Second, the choice of $v_{\text{ind}}(x)$ is heuristic, and better weight functions may improve performance.

Refinement by estimating $\bar{F}_{\boldsymbol{P}}$. To overcome these limitations, we refine the estimator by approximating $\bar{F}_{\boldsymbol{P}}$ instead of assuming it is negligible. Instead of estimating each $\boldsymbol{P}_t$ individually, we estimate the averaged alternative CDF $\bar{F}_{\boldsymbol{P}}$ using pivotal statistics collected from a comparable open-source model. Specifically, let $Y_1^{\text{wm}}, \ldots, Y_N^{\text{wm}}$ denote pivotal statistics derived from purely

watermarked texts, and define their empirical CDF as $\widehat{F}_{\boldsymbol{P}}(x) = \frac{1}{N}\sum_{t=1}^{N} \mathbf{1}\{Y_t^{\mathrm{wm}} \le x\}$. Since this estimation process can be performed prior to observing our samples, $\widehat{F}_{\boldsymbol{P}}$ is treated as a non-random CDF. Incorporating this additional information, the refined estimator is given by

$$\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) = \frac{F_0(\delta) - \widehat{F}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}. \tag{5}$$

Optimal weight function. Finally, we address the choice of weight function v. To gain some insights, let's consider an ideal case where $\bar{F}_{\boldsymbol{P}}$ is perfectly estimated such that $\widehat{F}_{\boldsymbol{P}} = \bar{F}_{\boldsymbol{P}}$. In this case, the variance of the resulting estimator $\frac{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\widehat{F}}[v]}{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]}$ for a given weight function v is upper bounded by (see Appendix A.3 for the proof)

$$\mathrm{Var}\left(\frac{\mathbb{E}_{F_0}[v] - \mathbb{E}_{\widehat{F}}[v]}{\mathbb{E}_{F_0}[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]}\right) \le \frac{\mathrm{Var}_{\bar{F}}(v)}{n(\mathbb{E}_{F_0}[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v])^2}. \tag{6}$$

Remark 3.4. *One can verify that* (6) *reduces to an exact equation when $Y_1, \ldots, Y_n$ are i.i.d. from $\bar{F}$. Under Model 2, where each $Y_t \sim (1-\varepsilon)F_0 + \varepsilon F_{\boldsymbol{P}_t}$, this condition holds if the $\boldsymbol{P}_t$ are also i.i.d. generated. Our minimax lower bound is established under this setting.*

We then minimize this variance upper bound to determine the optimal weight function.

Lemma 3.4 (Optimal weight function)**.** *For Inequality 6, the optimal value over all weight functions satisfies*

$$\min_v \frac{\mathrm{Var}_{\bar{F}}(v)}{[\mathbb{E}_{F_0}[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]]^2} = \left[\int \frac{[1-g(x)]^2}{(1-\varepsilon)+\varepsilon g(x)} \mathrm{d}F_0(x)\right]^{-1}$$

and the optimal solution (up to constant factors) is

$$v_{\mathrm{opt}}(x) = \frac{1-g(x)}{(1-\varepsilon)+\varepsilon g(x)}, \ \textit{with}\ g(x) = \frac{\mathrm{d}\bar{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}. \tag{7}$$

However, directly using this optimal weight function v_{opt} presents a challenge: it depends on both $\bar{F}_{\boldsymbol{P}}$ and ε, which are unknown. While $\bar{F}_{\boldsymbol{P}}$ can be estimated empirically by $\widehat{F}_{\boldsymbol{P}}$, substituting v_{opt} into (3) introduces ε into both sides of the equation, creating a self-referential problem. To resolve this, we define the refined estimator $\widehat{\varepsilon}_{\mathrm{opt}}$ as the fixed point of the following operator, that is, $\widehat{\varepsilon}_{\mathrm{opt}} = \widehat{\mathcal{T}}(\widehat{\varepsilon}_{\mathrm{opt}})$ and

$$\widehat{\mathcal{T}}(\varepsilon) = \mathcal{P}\left(\frac{\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}(x)\right]}{\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]}\right), \tag{8}$$

where $\mathcal{P}(x)$ projects ε onto the interval $[\varepsilon_{\min}, 1-\varepsilon_{\min}]$ to ensure numerical stability. Here, $\varepsilon_{\min}$ is a sufficiently small number (e.g., 10^{-3}) to ensure numerical stability and $\widehat{v}_{\mathrm{opt}}$ is the empirical version of the optimal weight function, defined as:

$$\widehat{v}_{\mathrm{opt}}(\varepsilon, x) = \frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}, \quad \widehat{g}(x) = \frac{\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}. \tag{9}$$

Here, $\widehat{g}(x)$ represents an empirical estimate of the density ratio function $g(x) = \frac{\mathrm{d}\bar{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}$, with its specific form detailed in the experiments section. Since $\widehat{g}(x)$ can be computed prior to the proportion estimation, we treat it as a non-random function in the theoretical analysis.

The intuition behind $\widehat{\varepsilon}_{\text{opt}}$ is that it balances two key factors: the weight function $\widehat{v}_{\text{opt}}$ leverages information about $\widehat{F}_{\boldsymbol{P}}$ to improve accuracy, while the fixed-point formulation ensures that the unknown ε is consistently estimated. This approach effectively incorporates both theoretical insights and empirical data to refine the estimation process. As Lemma 3.5 implies, $\widehat{\varepsilon}_{\text{opt}}$ generally exists and is unique because the operator is a contraction mapping.

Lemma 3.5. *Assume that* $0 < D_{\text{TV}}(F_0, \widehat{F}_{\boldsymbol{P}})$, $\sup_x |\widehat{g}(x)| < \infty$, *and* $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$. *If* $\sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}}$ *is sufficiently small and* n *is sufficiently large, then with high probability,* $\widehat{\mathcal{T}}$ *is a contraction mapping on the interval* $[\varepsilon_{\min}, 1-\varepsilon_{\min}]$ *for small values of* $\varepsilon_{\min}$.

4 Theoretical Analysis

In this section, we present a theoretical analysis of the estimators introduced earlier. Formal theories and detailed proofs for all results are provided in Appendix A.

4.1 Accuracy for Initial Attempts

Theorem 4.1. *Let* $\varepsilon \in [0,1]$ *and* $\delta \in (0,1]$ *be fixed. Under Model 2, it follows that*

$$\mathbb{E}\left|\widehat{\varepsilon}_{\text{ini}}(\delta) - \varepsilon + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}\right| \le \frac{\sigma_n}{\sqrt{n}} \quad \textit{where} \quad \sigma_n^2 = \frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta)]^2},$$

Corollary 2. *Following Theorem 4.1, we have*

$$\mathbb{E}\,|\widehat{\varepsilon}_{\text{ini}}(\delta_n) - \varepsilon| \le \frac{1}{\sqrt{nF_0(\delta_n)}} \cdot \sqrt{(1-\varepsilon) + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta_n)}{F_0(\delta_n)}} + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta_n)}{F_0(\delta_n)}.$$

Under Assumption 3.1, the right hand side $\to 0$ *if* $\delta_n \to 0$ *and* $nF_0(\delta_n) \to \infty$ *as* $n \to \infty$.

Theorem 4.1 shows that the mean absolute error (MAE) of $\widehat{\varepsilon}_{\text{ini}}(\delta)$, i.e., $\mathbb{E}\,|\widehat{\varepsilon}_{\text{ini}}(\delta) - \varepsilon|$, is of the order $\frac{\sigma_n}{\sqrt{n}}$ + bias. The noise term, $\frac{\sigma_n}{\sqrt{n}}$, arises from the randomness in the collected data. The bias term, $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}$, is due to neglecting the unknown term $\bar{F}_{\boldsymbol{P}}$ in the estimator $\widehat{\varepsilon}_{\text{ini}}$. In general, this bias cannot be eliminated, as we will show in Section 4.4 using the green-red list watermark as an example. However, for watermarks satisfying Assumption 3.1, this bias can be mitigated by setting δ sufficiently small, leveraging the condition $\lim_{\delta\to 0}\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)} = 0$, as show in Corollary 2. In summary, adaptively tuning δ can reduce the bias and make the estimator asymptotically unbiased.

4.2 Accuracy for Refinements

Theorem 4.2. *Let* $o_g(1)$ *(or* $o(1)$*) denote a quantity that converges to zero when* $\sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}}$ *(or* $\frac{1}{n}$*) converges to zero. Under Model 2, it follows that*

$$\mathbb{E}\,|\widehat{\varepsilon}_{\text{rfn}}(\delta) - \varepsilon| \le \frac{\sigma_n^\star}{\sqrt{n}} + o_g(1).$$

If $0 < D_{\mathrm{TV}}(F_0, \bar{F}_{\boldsymbol{P}})$, $\sup_x |g(x)| < \infty$ *and* $\mathbb{E}_{F_0}[g]^{-2} < \infty$, *then either setting* $\varepsilon_{\min}$ *as a small constant or choosing* $\varepsilon_{\min} = \frac{\log n}{\sqrt{n}}$, *we have that*

$$\mathbb{E}\,|\widehat{\varepsilon}_{\mathrm{opt}} - \varepsilon| \lesssim \frac{\tau_n^\star + o(1)}{\sqrt{n}} + o_g(1) + \varepsilon_{\min}\mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\}. \tag{10}$$

Here, $\sigma_n^\star \geq \tau_n^\star$ *are two variance terms defined by*

$$[\sigma_n^\star]^2 = \frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta) - \bar{F}_{\boldsymbol{P}}(\delta)]^2} \quad \text{and} \quad [\tau_n^\star]^2 = \left[\int \frac{[1-g(x)]^2}{(1-\varepsilon)+\varepsilon g(x)} \mathrm{d}F_0(x)\right]^{-1}. \tag{11}$$

Remark 4.3. *The error bound* (10) *remains valid when* $\mathbb{E}_{F_0}[g]^{-2} = \infty$ *and* $\varepsilon_{\min} = \frac{\log n}{n^{1/4}}$, *but achieving it requires a stronger condition:* $n^{1/4} \cdot \sup_x \frac{|\widehat{g}(x)-g(x)|}{\min 1, \widehat{g}(x)} \to 0$. *See Appendix A.6.2.*

Theorem 4.2 provides an analysis of the MAEs for the refined estimators $\widehat{\varepsilon}_{\mathrm{rfn}}(\delta)$ and $\widehat{\varepsilon}_{\mathrm{opt}}$. The analysis assumes that the density ratio estimate $\widehat{g}$ is sufficiently accurate, satisfying the condition $\sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}}$ being small. For $\widehat{\varepsilon}_{\mathrm{rfn}}(\delta)$, the MAE is given by $\frac{\sigma_n^\star}{\sqrt{n}}$. In contrast, analyzing the MAE for $\widehat{\varepsilon}_{\mathrm{opt}}$ requires additional moment conditions, which are necessary to study the fixed-point solutions of the operator $\widehat{\mathcal{T}}$ in (8). Under these conditions, $\widehat{\varepsilon}_{\mathrm{opt}}$ achieves an MAE of $\frac{\tau_n^\star}{\sqrt{n}}$, if we ignore higher-order terms and assume the stability constant $\varepsilon_{\min}$ serves as a valid lower bound for the true proportion ε. Note that $[\tau_n^\star]^2$ corresponds to the optimal value of the minimization problem in Lemma 3.4, as a result of which, $\sigma_n^\star \geq \tau_n^\star$. This guarantees that $\widehat{\varepsilon}_{\mathrm{opt}}$ consistently achieves a smaller MAE than $\widehat{\varepsilon}_{\mathrm{rfn}}(\delta)$, highlighting the advantage of using the optimal weight function. As will be shown in the experiments, $\widehat{\varepsilon}_{\mathrm{opt}}$ not only achieves better accuracy but also exhibits higher stability.

4.3 Minimax Lower Bounds

Theorem 4.4. *For a given* n, *let* $Y_1, \ldots, Y_n$ *be independently generated under Model 2, where* ε_0 *represents the true proportion, and the NTP distribution is uniformly sampled from the set* $\{\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n\}$. *Let* $\mathcal{B}(\varepsilon)$ *denote a small open interval containing a given proportion* ε. *Let* $[\sigma_n^\star]^2$ *and* $[\tau_n^\star]^2$ *be the variance term defined in* (11).

(*a*) *If* $\mathcal{G}$ *is the set of all measurable functions of the indicators* $\mathbf{1}\{Y_1 \leq \delta\}, \ldots, \mathbf{1}\{Y_n \leq \delta\}$,

$$\inf_{\widehat{\varepsilon} \in \mathcal{G}} \sup_{\varepsilon_0 \in \mathcal{B}(\varepsilon)} \mathbb{E}\,|\widehat{\varepsilon} - \varepsilon_0| \geq \frac{\sigma_n^\star}{\sqrt{2n}}.$$

(*b*) *If* $\mathcal{G}$ *is the set of all measurable functions of the random vector* $(Y_1, \ldots, Y_n)$,

$$\inf_{\widehat{\varepsilon} \in \mathcal{G}} \sup_{\varepsilon_0 \in \mathcal{B}(\varepsilon)} \mathbb{E}\,|\widehat{\varepsilon} - \varepsilon_0| \geq \frac{\tau_n^\star}{\sqrt{2n}}$$

Theorem 4.4 establishes minimax lower bounds for MAEs under Model 2, which characterizes the best achievable accuracy in a worst-case scenario. The worst case coincides with the setting where the $\boldsymbol{P}_t$'s are drawn i.i.d. from a large pool. While this may appear simplified, it reflects a realistic constraint: real-world verifiers lack access to exact NTP distributions and must rely on

aggregate statistics to refine estimation. In our formulation, these bounds (either $\frac{\sigma_n^\star}{\sqrt{n}}$ or $\frac{\tau_n^\star}{\sqrt{n}}$) are expressed in terms of the averaged CDF $\bar{F}_{\boldsymbol{P}}$. This CDF $\bar{F}_{\boldsymbol{P}}$ captures key distributional properties of the alternative distributions without requiring knowledge of individual $\boldsymbol{P}_t$'s, and it can be reasonably estimated using similar LLMs on related tasks. In contrast, more general bounds would require knowing which token corresponds to which $\boldsymbol{P}_t$, which is practically infeasible.

The implications are twofold. First, these lower bounds are tight and achieved by our estimators, up to the estimation error $o_g(1)$ in estimating $\bar{F}_{\boldsymbol{P}}$: $\widehat{\varepsilon}_{\mathrm{rfn}}(\delta)$ matches the bound $\frac{\sigma_n^\star}{\sqrt{n}}$ among indicator-based estimators, while $\widehat{\varepsilon}_{\mathrm{opt}}$ attains the smaller bound $\frac{\tau_n^\star}{\sqrt{n}}$ among all measurable estimators. Second, achieving these bounds critically depends on incorporating the correction term $\bar{F}_{\boldsymbol{P}}$; without it, the initial estimator fails to reach the optimal rate, as detailed in Appendix A.4.

4.4 Inherent Bias Caused by Non-identifiability

The green-red list watermark does not satisfy Assumption 3.1, so we cannot expect the aforementioned estimators to obtain high accuracy for this watermark. One piece of evidence is that, although Theorem 4.1 still holds for the green-red list watermark, the bias term $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)} = \varepsilon\left(1-\varepsilon+\varepsilon\frac{1-\bar{\mu}_t}{1-\gamma}\right)$ does not depend on δ and will never vanish as $\delta \to 0$. This is consistent with the non-identifiability issue: $\bar{\mu}_t$ is unobservable since each $\boldsymbol{P}_t$ is unobservable (see Table 1 for their relation), and the binary Y_t signals provide negligible information about $\bar{\mu}_t$. Even attempting to estimate $\bar{\mu}_t$ fails to resolve this issue. For instance, applying MLE to the i.i.d. binary mixture model in Lemma 3.1 to estimate unknown parameters μ and ε introduces inherent bias, as shown in Theorem 4.5.

Theorem 4.5. *Let $Y_1, \ldots, Y_n \overset{i.i.d.}{\sim} (1-\varepsilon)Ber(\gamma) + \varepsilon Ber(\mu)$, where only ε and μ $(1 > \mu > \gamma > 0)$ are unknown. Let $\widehat{e}$ denote the average of these n samples. Consider MLE that minimizes the L_2-regularized negative log-likelihood loss:*

$$(\widehat{\varepsilon}_\lambda, \widehat{\mu}_\lambda) = \arg\min \left\{ -\log L(\varepsilon, \mu) + \lambda(\mu^2 + \varepsilon^2) \right\}, \tag{12}$$

where $L(\varepsilon, \mu) = [(1-\varepsilon)\gamma + \varepsilon\mu]^{\widehat{e}}\,[1-(1-\varepsilon)\gamma - \varepsilon\mu]^{1-\widehat{e}}$. We then have the following results

$$\lim_{\lambda\to 0} \widehat{\varepsilon}_\lambda = x\sqrt{x^2+\gamma}, \quad \lim_{\lambda\to 0} \widehat{\mu}_\lambda = x^2 + \gamma, \tag{13}$$

for n sufficiently large where $x > 0$ is the solution to $x^3\sqrt{x^2+\gamma} = \widehat{e} - \gamma$.

As we will see in Figure 1e, the estimate $\widehat{\varepsilon}$ almost never equals the ground truth ε even when n is sufficiently large and λ is very small. This case is very different from standard linear regression, where regularized MLE (almost surely) converges to the underlying true parameter as regularization diminishes and sample size diverges [28]. This discrepancy arises because non-identifiability is an inherent problem of the (watermarking) model, not a flaw of the estimation methods.

5 Experiments

We first evaluate estimation accuracy and inherent bias on simulation datasets and then check performance on open-source LLMs. Additional details and results are in the appendix. The code is available at: `https://github.com/lx10077/WatermarkProportion`.

5.1 Simulation Study

Following [19], we use a vocabulary of size $|\mathcal{W}| = 10^3$ and generate 2,500 sequences of 400 tokens each, yielding $n = 10^6$ pivotal statistics as watermarked samples. The generation process mimics LLM behavior with a control parameter $\Delta \in (0, 1)$. For each token w_t, we construct its distribution $\boldsymbol{P}_t$ by uniformly sampling the largest probability from $U(0, 1 - \Delta)$ and interpolating between Zipf's law [46] and uniform distributions to ensure valid probabilities over $\mathcal{W}$ (Algorithm 1). We then sample $w_t \sim \boldsymbol{P}_t$. To simulate mixed-source data, we generate $n = 10^6$ null samples from F_0. Across 200 uniformly spaced values $\varepsilon \in [0, 1]$, we create mixture datasets combining $n(1 - \varepsilon)$ null samples with $n\varepsilon$ watermarked samples to estimate each ε. We aggregate all watermarked statistics into a single dataset for estimating $\widehat{g}$ and $\widehat{F}_{\boldsymbol{P}}$, though some pivotal statistics in this aggregated dataset may be entirely irrelevant to the current task of estimating a specific ε.

An equivalence for implementation. Since F_0 is known, the probability integral transform [9] ensures that $F_0(Y_1), \ldots, F_0(Y_n)$ are i.i.d. $U(0, 1)$ samples. We can therefore assume without loss of generality that $\mu_0 = U(0, 1)$, with $\mu_{1,\boldsymbol{P}} = F_{\boldsymbol{P}} \circ F_0^{-1}$ where F_0^{-1} is the generalized inverse. Assumption 3.1 remains valid under this transformation. For all experiments, we apply this transform to ensure $\mu_0 = U(0, 1)$, so our pivotal statistics are basically $F_0(Y_1), \ldots, F_0(Y_n)$. For simplicity, we continue denoting them as $Y_1, \ldots, Y_n$.

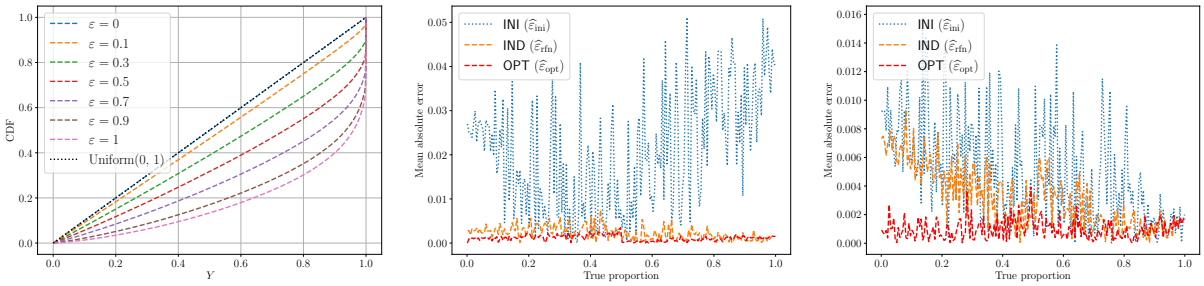
How we obtain $\widehat{g}$. The approach to estimating the density ratio g is highly flexible, with many methods available in the statistical literature. We estimate the density ratio g using a simple histogram-based approach. The data in $[0, 1]$ is divided into $N = 500$ bins, with bin heights normalized to form probability densities that integrate to 1, yielding a piecewise constant estimator. While more sophisticated methods like kernel density estimation could provide smoother estimates, they are computationally prohibitive for the large datasets ($n = 10^6$). The histogram approach offers better computational efficiency and scalability.

CDFs of mixture distributions. Figure 1a shows the CDFs of mixture sample distributions for the Gumbel-max watermark when $\Delta = 0.1$. As ε increases, the CDFs diverge further from the uniform distribution, which corresponds to $\varepsilon = 0$ (pure null samples). When $\varepsilon = 1$, the mixture consists entirely of watermarked samples, resulting in the greatest deviation from uniformity. Additional results for other values of Δ and the inverse transform watermark are provided in Figures 3 and 4 in the appendix.

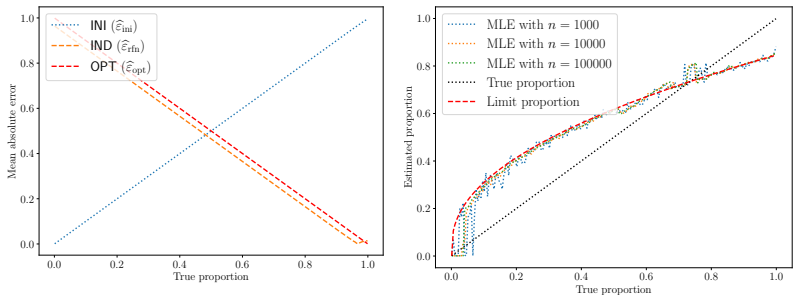
Estimation accuracy. We now examine the estimation accuracy, measured by the mean absolute errors (MAEs) on the $n = 10^6$ mixed-source samples. Three estimators are compared: (i) INI, the initial estimator $\widehat{\varepsilon}_{\text{ini}}(\delta)$ tuned with the best δ from $\{10^{-1}, 10^{-2}, 10^{-3}\}$, (ii) IND, the refined estimator $\widehat{\varepsilon}_{\text{rfn}}(\delta)$ with δ similarly tuned, and (iii) OPT, the refined estimator $\widehat{\varepsilon}_{\text{opt}}$ employing the optimal weight function $\widehat{v}_{\text{opt}}$ defined in (9).

Figures 1b and 1c visualize the MAEs of these estimators at various ground truth values ε when $\Delta = 0.1$. The results reveal that OPT consistently outperforms both INI and IND across all ε values, particularly at higher ε, where watermarked samples dominate. This shows the benefits of the optimal weight function $\widehat{v}_{\text{opt}}$ in improving estimation accuracy. While less accurate than OPT, IND still shows improvements over INI.

Figure 1: Figure 1a shows the CDFs of mixture sample distributions for the Gumbel-max watermark with $\Delta = 0.1$ and varying ε. Figures 1b, 1c and 1d show MAEs of three estimators across various ground truth ε values when $\Delta = 0.1$. Figure 1e shows the MLE results for estimating both ε and μ in the case of i.i.d. non-identifiable mixture binary data.



(a) CDFs of mixture dist. (b) MAEs for Gumbel-max (c) MAEs for inverse transform



(d) MAEs for green-red list (e) Inherent bias (green-red list)

Table 2: Averaged MAEs calculated over 200 ground truth ε values which are uniformly distributed on $[0, 1]$. Standard deviations are provided in parentheses, and all values are reported in units of 10^{-4}. Bold numbers denote the best performance.

Watermarks	Methods	$\Delta = 0.1$	$\Delta = 0.2$	$\Delta = 0.3$	$\Delta = 0.4$	$\Delta = 0.5$	$\Delta = 0.6$
Gumbel-max	INI	210(132)	140(99)	59(40)	95(60)	33(23)	26(13)
	IND	23(15)	22(14)	31(17)	13(9)	23(13)	20(13)
	OPT	**11(7)**	**13(8)**	**14(7)**	**9(5)**	**11(7)**	**13(7)**
Inverse	INI	51(36)	47(35)	16(11)	16(10)	13(10)	33(18)
	IND	32(22)	11(8)	**14(10)**	15(10)	13(10)	33(18)
	OPT	**10(7)**	**10(7)**	16(**10**)	**14(8)**	**11(7)**	**13(8)**

As further evidence, Table 2 presents the averaged MAEs across different Δ values. For both watermark types, the OPT estimator consistently achieves the lowest average MAEs with the smallest standard deviations, as highlighted by the bold numbers. This shows not only the accuracy of OPT but also its stability across varying conditions. For example, in the Gumbel-max watermark setting with $\Delta = 0.1$, OPT records an MAE of 11×10^{-4}, a significant improvement over INI (210×10^{-4}) and IND (23×10^{-4}). This performance pattern persists across other Δ values, showing the consistent superiority of OPT. The improvement of OPT over IND further implies the critical role of the optimal

weight function in enhancing estimation accuracy. This is expected, as the weight function is derived by minimizing the variance of the estimator (see Lemma 3.4), thus inherently providing robustness over diverse settings.

Non-identifiability. Our estimators are expected to fail on the green-red list watermark because ε is not identifiable in this case, given that NTP distributions $\boldsymbol{P}_{1:n}$ are inaccessible. A further issue is that binary $Y_{1:n}^{\mathrm{wm}} \equiv 1$, which results in $\widehat{F}_{\boldsymbol{P}} \equiv 1$ and provides no information about $\bar{F}_{\boldsymbol{P}}$, leading to abnormal estimation behavior. Figure 1d confirms this theoretical prediction: INI often gives zero estimates while IND and OPT give estimates near one.

Inherent bias. We then empirically validate the inherent bias predicted by Theorem 4.5 by generating n i.i.d. samples from $(1-\varepsilon)\mathrm{Ber}(\gamma) + \varepsilon\mathrm{Ber}(\mu)$, computing the L_2-regularized MLE via (12), and plotting $\widehat{\varepsilon}_\lambda$ against ε. Figure 1e shows results for $\lambda = 10^{-2}$ and $(\gamma, \mu) = (0.3, 0.9)$, where $\widehat{\varepsilon}_\lambda$ consistently deviates from true ε, confirming the predicted bias. The red dashed line representing the theoretical limit from (13) closely matches empirical estimates. Similar patterns across different parameter settings (Figure 7 in the appendix) show that non-identifiability renders standard estimation methods ineffective.

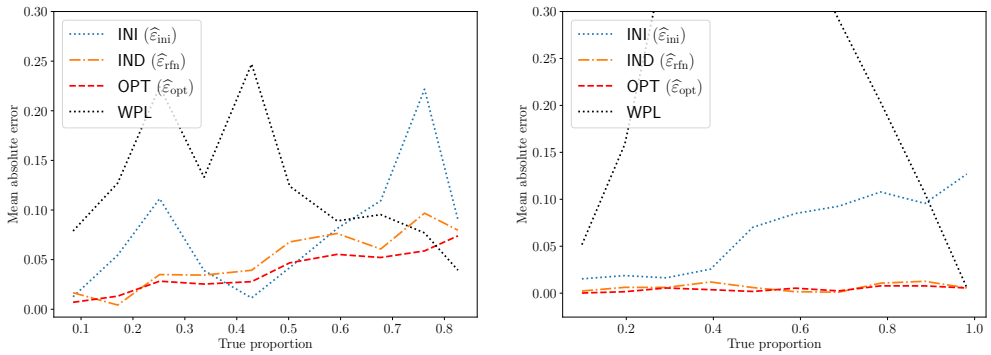
5.2 Open-source Model Experiments

We evaluate watermark proportion estimation using open-source LLMs under a controlled setup. We begin by sampling 500 documents from two corpora: the news-like C4 dataset [27] and the scientific arXiv dataset [6]. For each document, we extract the last 50 tokens as a prompt and generate 500 continuation tokens using one of three models: OPT-1.3B, OPT-13B [41], or LLaMA3.1-8B [11]. To introduce mixture data, at each step t, the LLM generates w_t using a specified watermarking scheme with probability ε; with the remaining probability, it samples w_t directly from $\boldsymbol{P}_t$ without watermarking. We further apply 1-sequence repeated context masking [8], which watermarks w_t only if the current context window $w_{(t-m):(t-1)}$ is unique in the generation history. This technique improves text quality but lowers the realized watermark proportion below the nominal ε, yielding a more realistic mix of watermarked and unwatermarked tokens. All experiments are conducted under two temperature settings $\mathrm{T} \in \{0.7, 1\}$. We adopt WPL [44] as our baseline since no existing method directly estimates watermark proportions.[2] WPL is a token-level localization method that assigns a binary label to each token indicating whether it is watermarked, and the overall proportion can be estimated by averaging these labels.

Estimation accuracy. MAE results for the 1.3B-OPT model and C4 dataset are shown in Figure 2, while results for other LLMs and datasets, with similar performance patterns, are provided in the appendix. Table 3 summarizes the mean and standard deviation of MAEs, offering an overview of estimation performance across models, temperatures, and watermarks. In nearly all cases, the optimal estimator OPT achieves the smallest error with the least variation, demonstrating strong empirical performance. In contrast, WPL performs much worse, with MAEs higher than any of our estimators. This result is expected, as token-level localization is inherently more challenging than estimating a global proportion, making WPL suboptimal for our task.

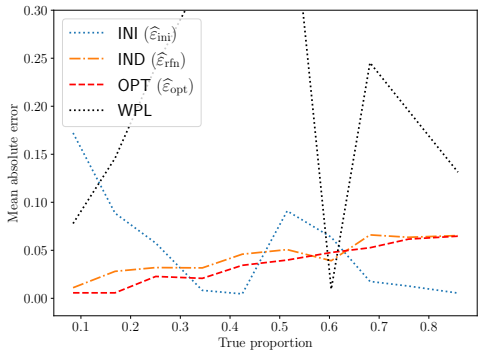
[2]We exclude another token-level localization method, SeedBS [18], due to its poor accuracy and prohibitive runtime.

Figure 2: MAEs of estimators across various true proportions ε and temperature parameters on the OPT-1.3B model and C4 dataset. Additional results for other models and datasets are provided in Appendix C.

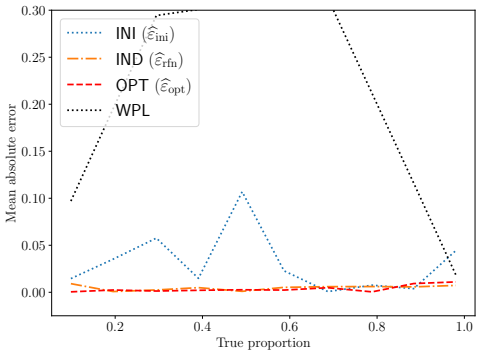


(a) Gumbel-max with T = 0.7

(b) Gumbel-max with T = 1



(c) Inverse with T = 0.7



(d) Inverse with T = 1

Table 3: Averaged MAEs calculated over 10 ground truth ε values on open-sources model experiments. Standard deviations are provided in parentheses, and all values are reported in units of 10^{-3}. Bold numbers denote the best performance.

Models	Datasets	T	Gumbel-max				Inverse transform			
			WPL	INI	IND	OPT	WPL	INI	IND	OPT
OPT-1.3B	C4	0.7	123(62)	77(59)	51(28)	**39(20)**	214(125)	52(52)	43(**17**)	**35**(21)
	C4	1	254(167)	65(40)	6(4)	**4(3)**	247(142)	31(31)	5(**3**)	**4(3)**
	Arxiv	1	275(184)	70(105)	19(8)	**11(6)**	286(174)	17(14)	18(**8**)	**12(8)**
OPT-13B	C4	0.7	119(90)	122(94)	34(19)	**26(15)**	212(135)	49(28)	25(**12**)	**20(12)**
	C4	1	195(156)	56(40)	8(5)	**5(3)**	250(143)	94(43)	70(48)	**68(40)**
	Arxiv	1	253(162)	51(27)	26(**9**)	**17**(10)	262(140)	27(15)	21(12)	**16(8)**
LLaMA-8B	C4	0.7	82(71)	60(34)	90(42)	**75(37)**	160(113)	60(34)	90(42)	**75(37)**
	C4	1	263(178)	45(22)	6(3)	**4(2)**	148(77)	30(32)	**2(1)**	5(4)
	Arxiv	1	236(201)	44(18)	18(9)	**15(7)**	291(176)	32(32)	19(11)	**16(6)**

Table 4: Average MAEs under common modifications, computed over 11 ground-truth ε values on the C4 dataset with temperature 1. Standard deviations are shown in parentheses, and all values are scaled by 10^{-3}. Boldface denotes the best performance.

Models	Edit types	Gumbel-max				Inverse transform			
		WPL	INI	IND	OPT	WPL	INI	IND	OPT
OPT-1.3B	Substitution	103(63)	56(24)	3(2)	**1(1)**	275(109)	46(35)	5(3)	**2(1)**
	Insertion	105(66)	70(35)	**8(5)**	**8(5)**	282(84)	38(23)	9(8)	**9(5)**
	Deletion	170(88)	**38**(27)	71(19)	66**(18)**	268(108)	35(27)	68(18)	**65(19)**
OPT-13B	Substitution	64(33)	78(55)	5(5)	**1(1)**	262(85)	52(28)	4(2)	**1(1)**
	Insertion	60(33)	61(32)	12**(4)**	8**(5)**	244(95)	50(40)	**8**(7)	**8(5)**
	Deletion	100(56)	**49**(38)	63**(15)**	66(18)	259(89)	26(23)	72**(17)**	68**(17)**
LLaMA-8B	Substitution	127(70)	54(28)	6(5)	**1(1)**	236(127)	50(37)	3(4)	**2(1)**
	Insertion	126(73)	46(22)	7(4)	**2(1)**	243(121)	18(15)	8(6)	**3(2)**
	Deletion	201(60)	**34**(22)	56(24)	38**(20)**	270(109)	54(47)	54(21)	**36(19)**

Performance after modification. We then evaluate the performance of proportion estimators under post-processing modifications. Starting from a fully watermarked dataset, we modify a given proportion of tokens using random substitution, deletion, or insertion, and estimate the resulting watermark proportion. The true proportion is defined as the fraction of pivotal statistics following the alternative distribution. Specifically, if a token w_t or any of its m preceding tokens is modified, we treat Y_t as null. Thus, modifying an ε-fraction of tokens may affect up to an $m\varepsilon$-fraction of pivotal statistics, resulting in a watermark proportion of at least $1 - m\varepsilon$. Results on the C4 dataset are shown in Table 4, with similar findings on the arXiv dataset in Table 5 in the appendix.

We have two observations. First, our estimators remain robust under post-processing modifications, with OPT achieving the lowest MAE and variance in most cases, though INI or IND may occasionally perform better. Second, all estimators perform slightly better than in the mixture setting (Table 3). This improvement arises because, in the post-processing setup, only a subset of tokens is modified, while the rest remain fully watermarked. This results in longer uninterrupted spans of watermarked text, making the underlying signal easier to detect than in the mixture setting, where watermarked and non-watermarked tokens are interleaved throughout. This improvement also underscores the broad applicability of our formulation in Section 3.1.

6 Discussion

In this paper, we address the problem of estimating the proportion of watermarked pivotal statistics in mixed-source texts. We formulate this task as a coefficient estimation problem in a time-varying mixture model and establish conditions under which the proportion is identifiable. For the green-red list watermark, the proportion is not identifiable due to the binary nature of its pivotal statistic, which provides limited information. As a result, common estimation methods, such as maximum likelihood estimation, suffer from inherent and unavoidable bias. In contrast, for watermarks with continuous pivotal statistics, such as the Gumbel-max and inverse transform watermarks, we demonstrate that the proportion is identifiable. We propose several efficient estimators and show that they achieve the minimax optimal lower bounds within their respective classes. Our methods exhibit strong

performance on both synthetic datasets and LLM-generated data.

Several open directions remain for future research. This work measures the extent of AI-generated content through pivotal statistics but does not account for its importance within mixed-source texts. For instance, 10% AI-generated content might carry critical significance in some contexts but be less relevant in others. Incorporating additional information, such as token semantics or entropies, could enable more nuanced proportion estimation. Exploring how to integrate these features is a promising avenue for future study. Another direction is to investigate the theoretical limits of identifying AI-generated segments, including possible phase transitions, similar to classic work [10, 5]. Developing efficient and adaptive methods for segment identification could further advance the field. Additionally, while our paper focuses on scalar pivotal statistics, some watermarking schemes, such as [8], use high-dimensional pivotal statistics. It even requires training neural networks for detection due to their complexity. Estimating the watermark proportion in such high-dimensional settings remains an open challenge. Addressing these issues could significantly enhance our understanding and capabilities in watermark proportion estimation.

Acknowledgments

We would like to thank two anonymous referees for their constructive comments that helped improve the presentation of the paper. This work was supported in part by NIH grants, RF1AG063481 and U01CA274576, NSF DMS-2310679, a Meta Faculty Research Award, and Wharton AI for Business. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

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Appendices

A Proofs of Theoretical Analysis

A.1 Proof of Results in Section 3.2

We provide the proofs of Lemmas 3.1, 3.2, and 3.3 in this subsection.

Proof of Lemma 3.1. Each data point is equivalently i.i.d. from $\text{Ber}((1-\varepsilon)\gamma+\varepsilon\mu)$. Since both ε and μ are unknown, there are multiple combinations of (ε, μ) that yield the same value for $(1-\varepsilon)\gamma+\varepsilon\mu$, resulting in the same data distribution. □

Proof of Lemma 3.2. By Assumption 3.1, we have three key conditions: (i) F_0 is known, (ii) $\lim_{x\to 0} \frac{\bar{F}(x)}{F_0(x)}$ is determined by the population distribution of observed Y, and (iii) $\lim_{x\to 0} \frac{\bar{F}_{\boldsymbol{P}}(x)}{F_0(x)} = 0$ since all NTP distributions are non-singular.

From (2), for any x,

$$\bar{F}(x) = (1-\varepsilon)F_0(x) + \varepsilon\bar{F}_{\boldsymbol{P}}(x). \tag{2}$$

Dividing both sides by $F_0(x)$ and taking the limit as $x \to 0$:

$$\lim_{x\to 0} \frac{\bar{F}(x)}{F_0(x)} = (1-\varepsilon) + \varepsilon \cdot \lim_{x\to 0} \frac{\bar{F}_{\boldsymbol{P}}(x)}{F_0(x)} = (1-\varepsilon) + \varepsilon \cdot 0 = 1-\varepsilon.$$

Therefore,

$$\varepsilon = 1 - \lim_{x\to 0} \frac{\bar{F}(x)}{F_0(x)}.$$

Since the right-hand side consists only of known or determined quantities, we conclude that ε is identifiable. □

Proof of Lemma 3.3. It suffices to prove that the null and alternative CDFs (as shown in Table 1) of the considered watermarks satisfy Assumption 3.1. We examine each watermark separately.
Gumbel-max watermark: The CDFs are given by

$$F_0(x) = x, \quad \text{and} \quad F_{\boldsymbol{P}}(x) = \sum_w P_w x^{1/P_w}.$$

Since $\boldsymbol{P} = (P_1, \ldots, P_{|\mathcal{W}|})$ is non-singular, we have $1/P_w > 1$ for all $w \in \mathcal{W}$. Therefore,

$$F_0'(0) = 1 > F_{\boldsymbol{P}}'(0) = 0 \quad \Longrightarrow \quad \lim_{x\to 0} \frac{F_{\boldsymbol{P}}(x)}{F_0(x)} = 0.$$

Inverse transform watermark: Following Li et al. [20], we consider the condition $P_{(2)} \cdot \log|\mathcal{W}| \to 0$ as $|\mathcal{W}| \to \infty$, where $P_{(i)}$ denotes the i-th largest probability in $\boldsymbol{P}$. The CDFs are:

$$F_0(x) = x^2, \quad \text{and} \quad F_{\boldsymbol{P}}(x) = \left[1 - \frac{1-x}{P_{(1)}}\right]_{[0,1]}^2,$$

where $[x]_{[0,1]} := \max\{\min\{x, 1\}, 0\}$. For any non-singular $\boldsymbol{P}$ (i.e., where 1 is not an entry of $\boldsymbol{P}$), we can verify that:

$$\frac{F_{\boldsymbol{P}}(x)}{F_0(x)} = 0 \ \ \forall x \in (0, 1 - P_{(1)}) \quad \Longrightarrow \quad \lim_{x \to 0} \frac{F_{\boldsymbol{P}}(x)}{F_0(x)} = 0.$$

Thus, both watermarks satisfy Assumption 3.1. □

A.2 Proof of Lemma 3.4

Proof of Lemma 3.4. We aim to maximize the following functional

$$\mathcal{L}(v) = \frac{\left(\mathbb{E}_0[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]\right)^2}{\mathrm{Var}_{\bar{F}}(v)},$$

where $v(x)$ is the function to optimize and $v_{\mathrm{opt}}(\varepsilon, x)$ is defined as:

$$v_{\mathrm{opt}}(\varepsilon, x) = \frac{\mathrm{d}F_0}{\mathrm{d}\bar{F}}(x) - \frac{\mathrm{d}\bar{F}_{\boldsymbol{P}}}{\mathrm{d}\bar{F}}(x).$$

In the main text, we use the equivalent form of $v_{\mathrm{opt}}(\varepsilon, x)$:

$$v_{\mathrm{opt}}(\varepsilon, x) = \frac{1 - g(x)}{(1 - \varepsilon) + \varepsilon g(x)} \quad \text{where} \quad g(x) = \frac{\mathrm{d}\bar{F}_{\boldsymbol{P}}}{\mathrm{d}F_0}(x).$$

The first step is to simplify the expression for $\mathcal{L}(v)$. Using the definition of $v_{\mathrm{opt}}(\varepsilon, x)$, its numerator becomes:

$$\mathbb{E}_0[v] - \mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v] = \int v(x) \left(\mathrm{d}F_0(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x)\right) = \int v(x) v_{\mathrm{opt}}(\varepsilon, x) \mathrm{d}\bar{F}(x).$$

Next, we analyze the denominator. Assume, without loss of generality, that $v(x)$ is centered under $\bar{F}$, i.e., $\int v(x)\mathrm{d}\bar{F}(x) = 0$. This is because adding a constant does not affect the value of $\mathcal{L}(v)$; that is, $\mathcal{L}(v) = \mathcal{L}(v + C)$ for any constant C. Then:

$$\mathrm{Var}_{\bar{F}}(v) = \int v^2(x)\mathrm{d}\bar{F}(x) - \left(\int v(x)\mathrm{d}\bar{F}(x)\right)^2 = \int v^2(x)\mathrm{d}\bar{F}(x).$$

Thus, the objective simplifies to:

$$\mathcal{L}(v) = \frac{\left(\int v(x) v_{\mathrm{opt}}(\varepsilon, x)\mathrm{d}\bar{F}(x)\right)^2}{\int v^2(x)\mathrm{d}\bar{F}(x)}.$$

By the Cauchy-Schwarz inequality, for two functions $v(x)$ and $v_{\mathrm{opt}}(\varepsilon, x)$ under the measure $\bar{F}$, we have:

$$\left(\int v(x) v_{\mathrm{opt}}(\varepsilon, x)\mathrm{d}\bar{F}(x)\right)^2 \leq \left(\int v^2(x)\mathrm{d}\bar{F}(x)\right)\left(\int v_{\mathrm{opt}}^2(\varepsilon, x)\mathrm{d}\bar{F}(x)\right).$$

Equality holds if and only if $v(x)$ is proportional to $v_{\mathrm{opt}}(\varepsilon, x)$, i.e. for some constant c,

$$v(x) = c \cdot v_{\mathrm{opt}}(\varepsilon, x).$$

Finally, plugging this optimal weight function, we have that

$$\mathcal{L}(v_{\mathrm{opt}}) = \frac{\left(\int v_{\mathrm{opt}}^2(\varepsilon, x)\mathrm{d}\bar{F}(x)\right)^2}{\int v_{\mathrm{opt}}^2(\varepsilon, x)\mathrm{d}\bar{F}(x)} = \int \frac{(\mathrm{d}F_0(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x))^2}{\mathrm{d}\bar{F}(x)} = \int \frac{(1 - g(x))^2}{(1 - \varepsilon) + \varepsilon g(x)} \mathrm{d}F_0(x).$$

□

A.3 Proof of Inequality (6)

Proof of Inequality (6). Finally, we prove the following inequality

$$\mathrm{Var}\left(\frac{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\widehat{F}}[v]}{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]}\right) \le \frac{\mathrm{Var}_{\bar{F}}(v)}{n(\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v])^2}. \tag{6}$$

Recall that under Model 2, $\widehat{F}$ is the empirical CDF of n independent pivotal statistics $Y_1,\ldots,Y_n$ where each $Y_t \sim (1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}$. It then follows that

$$\begin{aligned}\mathrm{Var}\left(\frac{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\widehat{F}}[v]}{\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v]}\right) &= \frac{\mathrm{Var}(\mathbb{E}_{\widehat{F}}[v])}{(\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v])^2} = \frac{\mathrm{Var}(\frac{1}{n}\sum_{t=1}^n v(Y_t))}{(\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v])^2}\\ &= \frac{\sum_{t=1}^n \mathrm{Var}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}(v)}{n^2(\mathbb{E}_{F_0}[v]-\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[v])^2}.\end{aligned}$$

Using the notation $\bar{F}=(1-\varepsilon)F_0+\varepsilon\cdot\frac{1}{n}\sum_{t=1}^n F_{\boldsymbol{P}_t}$, we then complete the proof by

$$\begin{aligned}\frac{1}{n}\sum_{t=1}^n \mathrm{Var}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}(v) &= \frac{1}{n}\sum_{t=1}^n [\mathbb{E}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}[v^2]-(\mathbb{E}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}[v])^2]\\ &\le \frac{1}{n}\sum_{t=1}^n \mathbb{E}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}[v^2]-\left(\frac{1}{n}\sum_{t=1}^n \mathbb{E}_{(1-\varepsilon)F_0+\varepsilon F_{\boldsymbol{P}_t}}[v]\right)^2\\ &= \mathbb{E}_{\bar{F}}[v^2]-(\mathbb{E}_{\bar{F}}[v])^2 = \mathrm{Var}_{\bar{F}}(v).\end{aligned}$$

□

A.4 Analysis for the Initial Estimator

We first present the proof of Theorem 4.1 and then discuss the MAE associated with adaptively tuning the parameter δ.

Proof of Theorem 4.1. Note that $\bar{F}=(1-\varepsilon)F_0+\varepsilon\bar{F}_{\boldsymbol{P}}$. We have the following decomposition:

$$\widehat{\varepsilon}_{\mathrm{ini}}(\delta) = 1-\frac{\widehat{F}(\delta)}{F_0(\delta)} = \frac{\bar{F}(\delta)-\widehat{F}(\delta)}{F_0(\delta)} - \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}+\varepsilon.$$

We denote by $p_t(\varepsilon)=(1-\varepsilon)F_0(\delta)+\varepsilon F_{\boldsymbol{P}_t}(\delta)$ for simplicity. Due to the relation (2), we have $\bar{F}(\delta)=\frac{1}{n}\sum_{t=1}^n p_t(\varepsilon)$. It then follows that

$$\begin{aligned}\mathbb{E}\left|\widehat{\varepsilon}_{\mathrm{ini}}(\delta)-\varepsilon+\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}\right| &\overset{(a)}{\le} \sqrt{\mathbb{E}\left|\widehat{\varepsilon}_{\mathrm{ini}}(\delta)-\varepsilon+\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}\right|^2}\\ &= \sqrt{\mathbb{E}\left|\frac{\bar{F}(\delta)-\widehat{F}(\delta)}{F_0(\delta)}\right|^2}\\ &= \frac{1}{F_0(\delta)n}\sqrt{\sum_{t=1}^n p_t(\varepsilon)(1-p_t(\varepsilon))}\end{aligned}$$

$$\overset{(b)}{\leq} \frac{1}{F_0(\delta)\sqrt{n}} \cdot \sqrt{\left[\frac{1}{n}\sum_{t=1}^{n} p_t(\varepsilon)\right]\left(1 - \left[\frac{1}{n}\sum_{t=1}^{n} p_t(\varepsilon)\right]\right)}$$
$$\overset{(c)}{=} \frac{\sigma_n}{\sqrt{n}},$$

where (a) and (b) follow from Jensen's inequality, and (c) utilizes the definition of σ_n that is $\sigma_n^2 = \frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta)]^2}$. □

Theorem 4.1 analyzes the mean absolute error (MAE) of $\widehat{\varepsilon}_{\text{ini}}(\delta)$ and shows it is of the order $\frac{\sigma_n}{\sqrt{n}} + \text{bias}$, where the bias term is given by $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}$. Theorem 4.4 establishes that the minimax optimal MAE within the class of all measurable functions of the indicator variables $\mathbf{1}\{Y_1 \leq \delta\}, \ldots, \mathbf{1}\{Y_n \leq \delta\}$ is $\frac{\sigma_n^\star}{\sqrt{n}}$. By Theorem 4.1, the variance term σ_n for $\widehat{\varepsilon}_{\text{ini}}(\delta)$ is given by $\frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta)]^2}$. As shown in Proposition A.1, this variance is smaller (up to constant factors) than the optimal variance $\sigma_n^\star$. However, the bias term $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)}$ generally increases the MAE of $\widehat{\varepsilon}_{\text{ini}}(\delta)$, making its overall MAE larger than $\frac{\sigma_n^\star}{\sqrt{n}}$. This is further detailed in Corollary 3.

Proposition A.1. *Under Assumption 3.1, we have $c \cdot \sigma_n \leq \sigma_n^\star$ for some $c > 0$, where*

$$\sigma_n^2 = \frac{\bar{F}(\varepsilon)(1 - \bar{F}(\varepsilon))}{[F_0(\delta)]^2}.$$

Proof of Proposition A.1. We observe that under Assumption 3.1, $\lim_{\delta\to 0} \frac{F_{\boldsymbol{P}}(\delta)}{F_0(\delta)} = 0$ ensuring that there exists a positive constant $c_1 > 0$ such that $|F_0(\delta) - F_{\boldsymbol{P}}(\delta)| \leq cF_0(\delta)$ for any $\boldsymbol{P}$, due to the continuity of $F_{\boldsymbol{P}}$ in $\boldsymbol{P}$. It implies that we have

$$\sigma_n^2 = \frac{\bar{F}(\delta)(1 - \bar{F}(\delta))}{[F_0(\delta)]^2} \leq c^2\frac{\bar{F}(\delta)(1 - \bar{F}(\delta))}{[F_0(\delta) - \bar{F}_{\boldsymbol{P}}(\delta)]^2} = c^2 \cdot [\sigma_n^\star]^2.$$

□

Corollary 3. *Under the same setting as Theorem 4.1, assume that $\bar{F}_{\boldsymbol{P}}(x) \sim x^p$ and $F_0(x) \sim x$ as $x \to 0$, where $p > 1$. Then, if $\varepsilon = \Theta(1)$, we have:*

$$\mathbb{E}\,|\widehat{\varepsilon}_{\text{ini}}(\delta_n) - \varepsilon| \leq \frac{\sigma_n}{\sqrt{n}} + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta_n)}{F_0(\delta_n)} \sim \frac{1}{\sqrt{n\delta_n}} + \varepsilon\delta_n^{p-1}.$$

By selecting the optimal value $\delta_n = [2(p-1)\varepsilon\sqrt{n}]^{-\frac{1}{p-0.5}}$, it follows that:

$$\mathbb{E}\,|\widehat{\varepsilon}_{\text{ini}}(\delta_n) - \varepsilon| \precsim \frac{\varepsilon^{\frac{1}{p-0.5}}}{n^{\frac{1}{2}\left(1-\frac{1}{p-0.5}\right)}}.$$

This result highlights that the bias term $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta_n)}{F_0(\delta_n)}$ limits the performance of $\widehat{\varepsilon}_{\text{ini}}(\delta_n)$. Even with adaptive tuning of δ, its dependence on n is strictly slower than $\frac{1}{\sqrt{n}}$, illustrating why $\widehat{\varepsilon}_{\text{ini}}(\delta)$ cannot achieve minimax optimality.

A.5 Analysis for the Refined Estimator

Theorem A.1 (Full version of the first part in Theorem 4.2)**.** *Let $\varepsilon \in [0,1]$ and $\delta \in (0,1]$ be fixed. Then,*

$$\mathbb{E}\left|\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) - \varepsilon + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}\right| \le \frac{\widehat{\sigma}_n}{\sqrt{n}}, \quad \textit{where} \;\; \widehat{\sigma}_n^2 = \frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)]^2}.$$

Corollary 4 (Simplified MAE)**.** *When $\sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}} \to 0$, it follows that $\widehat{\sigma}_n \to \sigma_n^\star$, implying $\sigma_n^\star = \widehat{\sigma}_n + o_g(1)$. Under this condition, the bias term $\varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta)-\widehat{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta)-\widehat{F}_{\boldsymbol{P}}(\delta)}$ also becomes $o_g(1)$. Consequently, we have*

$$\mathbb{E}\,|\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) - \varepsilon| \le \frac{\sigma_n^\star + o_g(1)}{\sqrt{n}}.$$

Proof of Theorem A.1. Note that $\bar{F} = (1-\varepsilon)F_0 + \varepsilon\bar{F}_{\boldsymbol{P}}$. We have the following decomposition:

$$\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) = \frac{F_0(\delta) - \widehat{F}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)} = \frac{\bar{F}(\delta) - \widehat{F}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)} - \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)} + \varepsilon.$$

We denote by $p_t(\varepsilon) = (1-\varepsilon)F_0(\delta) + \varepsilon F_{\boldsymbol{P}_t}(\delta)$ for simplicity. Due to the relation (2), we have $\bar{F}(\delta) = \frac{1}{n}\sum_{t=1}^n p_t(\varepsilon)$. It then follows that

$$\begin{aligned}
\mathbb{E}\left|\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) - \varepsilon + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}\right| &\overset{(a)}{\le} \sqrt{\mathbb{E}\left|\widehat{\varepsilon}_{\mathrm{rfn}}(\delta) - \varepsilon + \varepsilon\frac{\bar{F}_{\boldsymbol{P}}(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}\right|^2} \\
&= \sqrt{\mathbb{E}\left|\frac{\bar{F}(\delta) - \widehat{F}(\delta)}{F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)}\right|^2} \\
&= \frac{\sqrt{\sum_{t=1}^n p_t(\varepsilon)(1-p_t(\varepsilon))}}{n[F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)]} \\
&\overset{(b)}{\le} \frac{\sqrt{\left[\frac{1}{n}\sum_{t=1}^n p_t(\varepsilon)\right]\left(1 - \left[\frac{1}{n}\sum_{t=1}^n p_t(\varepsilon)\right]\right)}}{\sqrt{n}[F_0(\delta) - \widehat{F}_{\boldsymbol{P}}(\delta)]} \\
&\overset{(c)}{=} \frac{\widehat{\sigma}_n}{\sqrt{n}},
\end{aligned}$$

where (a) and (b) follow from Jensen's inequality, and (c) utilizes the definition of $\widehat{\sigma}_n$ that is $\widehat{\sigma}_n^2 = \frac{\bar{F}(\delta)(1-\bar{F}(\delta))}{[F_0(\delta)-\widehat{F}_{\boldsymbol{P}}(\delta)]^2}$. □

A.6 Analysis for the Optimal Estimator

A.6.1 Proof of Lemma 3.5

Recall that we define an operator $\widehat{\mathcal{T}} : [\varepsilon_{\min}, 1-\varepsilon_{\min}] \to [\varepsilon_{\min}, 1-\varepsilon_{\min}]$ in defining the optimal estimator $\widehat{\varepsilon}_{\mathrm{opt}}$:

$$\widehat{\mathcal{T}}(\varepsilon) = \mathcal{P} \circ \widetilde{\mathcal{T}}(\varepsilon) \;\; \text{with} \;\; \widetilde{\mathcal{T}}(\varepsilon) := \frac{\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}(x)\right]}{\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]}, \tag{14}$$

where $\mathcal{P}(x) := \max\{\min\{x, 1-\varepsilon_{\min}\}, \varepsilon_{\min}\}$ is the projection operator, and $\varepsilon_{\min} > 0$ is a small constant to avoid numerical issues.

To establish Lemma 3.5, which asserts that $\widehat{\mathcal{T}}$ is a contraction mapping with high probability, we first derive a supporting result, stated as Lemma A.1. The proof of Lemma A.1 is deferred to Appendix A.6.3.

Lemma A.1. *Define the total variation distance between F_0 and $\widehat{F}_{\boldsymbol{P}}$ as $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) := \frac{1}{2}\int |1 - \widehat{g}(x)| \mathrm{d}F_0(x)$ and $C_g := \max\{1, \sup_{x\in[0,1]} |\widehat{g}(x)|\}$. Assume that $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) > 0$ and $C_g < \infty$. Define*

$$\Delta_g = \sup_{x\in[0,1]} \frac{|\widehat{g}(x) - g(x)|}{\min\{1, \widehat{g}(x)\}} \quad \text{and} \quad C_0 = \frac{C_g^2}{2D_{\mathrm{TV}}^3(F_0, \widehat{F}_{\boldsymbol{P}})}.$$

Then, for some positive constant $K > 0$, the following results hold:

$$\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right| \geq \frac{C_0}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{n}} \right) \leq \frac{8K}{\varepsilon_{\min}^3} \cdot \frac{1}{n^{C-1}}, \tag{15a}$$

$$\sup_{\varepsilon\in[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right| \leq \Delta_g \left[1 + C_0 \min\left\{ \frac{1}{\varepsilon_{\min}}, \mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2} \right\} \right]. \tag{15b}$$

If we further have $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$, then (15a) *can be refined to*

$$\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right| \geq C_0' \left[\frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}} \right] \right) \leq \frac{8K}{\varepsilon_{\min}^3} \cdot \frac{1}{n^{C-1}}. \tag{15c}$$

where

$$C_0' := \frac{C_g^2\left[\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}\right]}{D_{\mathrm{TV}}^4(F_0, \widehat{F}_{\boldsymbol{P}})} \quad \text{and} \quad M := 8\,\mathbb{E}_{\bar{F}}\left[\widehat{g} - 1/\widehat{g}\right]^2.$$

Lemma 3.5 is a direct corollary of Lemma A.1. Another useful consequence of Lemma A.1 is that it guarantees, with high probability, the quantity $\sup_{\varepsilon\in[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \frac{\mathrm{d}\widehat{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right|$ is bounded by a small value A (see Corollary 5).

Corollary 5 (Full version of Lemma 3.5)**.** *Under the same assumptions and notations in Lemma A.1, for any $C > 0$, with probability at least $1 - \frac{8K}{\varepsilon_{\min}^3} \cdot \frac{1}{n^{C-1}}$,*

$$\sup_{\varepsilon\in[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \frac{\mathrm{d}\widehat{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right| \leq A,$$

where the bound A takes the following form depending on whether a moment condition holds:

- *If $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ (so that $\mathbb{E}_{F_0}[\widehat{g}]^{-2} < \infty$ also holds), the bound A is given by*

$$A = \Delta_g\left[1 + C_0\left(\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}\right)\right] + C_0'\left[\frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}\right].$$

- *If the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ fails, the bound A becomes*

$$A = \Delta_g \left[1 + \frac{C_0}{\varepsilon_{\min}}\right] + \frac{C_0}{\varepsilon_{\min}^2}\sqrt{\frac{C \log n}{n}}.$$

Proof of Corollary 5. Note that the projection operator enforces the derivative to be zero only when its output lies outside the domain $[\varepsilon_{\min}, 1-\varepsilon_{\min}]$. Therefore, we have

$$\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\frac{\mathrm{d}\widehat{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \le \sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right|.$$

By (15b) in Lemma A.1, it follows that or any $x > 0$,

$$\begin{aligned}
&\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \ge x + \Delta_g\left[1 + \frac{C_0}{4}\min\left\{\frac{1}{\varepsilon_{\min}^2}, \mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}\right\}\right]\right) \\
&\le \mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| - \sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \ge x\right) \\
&\le \mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]} \left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \ge x\right).
\end{aligned}$$

By setting $x = \frac{C_0}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{n}}$ and applying (15a), we complete the proof of the second bullet point. The proof of the first bullet point is similar and is therefore omitted. □

A.6.2 Proof of Theorem 4.2

The first part of Theorem 4.2 concerns the accuracy of the refined estimator $\widehat{\varepsilon}_{\mathrm{rfn}}$, with its analysis provided in Theorem A.1. The second part addresses the accuracy of the optimal estimator $\widehat{\varepsilon}_{\mathrm{opt}}$, which is analyzed in Theorem A.2. The inequality that $\sigma_n^\star \ge \tau_n^\star$ is established at the end of Appendix A.8.

Theorem A.2 (Full version of the second part of Theorem 4.2; proof in Appendix A.6.4)**.** *Assume $0 < D_{\mathrm{TV}}(\widehat{F}_{\boldsymbol{P}}, F_0)$ and $\sup_x |\widehat{g}(x)| < \infty$. Let $C_g := \max\{1, \sup_{x\in[0,1]} \widehat{g}(x)\}$. Define*

$$\Delta_g = \sup_{x\in[0,1]} \frac{|\widehat{g}(x) - g(x)|}{\min\{1, \widehat{g}(x)\}} \quad \textit{and} \quad C_0 = \frac{C_g^2}{D_{\mathrm{TV}}^3(F_0, \widehat{F}_{\boldsymbol{P}})}.$$

For a fixed true proportion $\varepsilon \in [0,1]$, it follows that

$$\mathbb{E}\,|\widehat{\varepsilon}_{\mathrm{opt}} - \varepsilon| \le (1-A)^{-1}\left(\mathcal{V} + \mathcal{B} + \varepsilon_{\min}\mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\}\right) + \frac{8K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

where $C > 0$ is any given positive constant, the expression of A is given in Corollary 5 (not restated here for brevity), K, C_0 are problem-dependent constants defined in Lemma A.1, and $\mathcal{V}$ and $\mathcal{B}$ are the variance and bias terms, respectively, defined as

$$\mathcal{V} = \frac{\sqrt{\frac{1}{n}\int\left[\frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\right]^2 \mathrm{d}\bar{F}(x)}}{\int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\mathrm{d}F_0(x)}, \quad \textit{and} \quad \mathcal{B} = \varepsilon\left|\frac{\int \widehat{v}_{\mathrm{opt}}(\varepsilon,x)\left[\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x)\right]}{\int \widehat{v}_{\mathrm{opt}}(\varepsilon,x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]}\right|.$$

Corollary 6 (Simplified MAE). *Assume* $0 < D_{\mathrm{TV}}(\bar{F}_{\boldsymbol{P}}, F_0)$, $\sup_x |g(x)| < \infty$, *and* $\mathbb{E}_{F_0}[g]^{-2} < \infty$. *Let* $o_g(1)$ *denote a term that converges to zero as* $\Delta_g := \sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}} \to 0$, *and* $o(1)$ *a term that converges to zero as* $n \to \infty$. *If either setting* $\varepsilon_{\min}$ *as a small constant or choosing* $\varepsilon_{\min} = \frac{\log n}{\sqrt{n}}$, *we then have*

$$A = o_g(1) + o(1), \quad \mathcal{B} = o_g(1), \quad \mathcal{V} = \frac{\tau_n^\star + o_g(1)}{\sqrt{n}}.$$

As a result, it follows that

$$\mathbb{E}\,|\widehat{\varepsilon}_{\mathrm{opt}} - \varepsilon| \precsim \frac{\tau_n^\star + o(1)}{\sqrt{n}} + o_g(1) + \varepsilon_{\min}\mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\}.$$

The last inequality also holds without the condition $\mathbb{E}_{F_0}[g]^{-2} < \infty$ *given that we set* $\varepsilon_{\min} = \frac{\log n}{n^{1/4}}$ *and ensure that* $n^{1/4} \cdot \Delta_g = o(1)$.

Proof of Corollary 6. It suffices to simplify the variance term $\mathcal{V}$ and the bias term $\mathcal{B}$ defined in Theorem A.2. Let $\Delta_g = \sup_x \frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}}$ denote the uniform estiamtion error between $\widehat{g}$ and g. Note that $\mathcal{V}$ is a continuous functional of the density ratio $\widehat{g}$. Hence, as $\Delta_g \to 0$, it follows that

$$\mathcal{V} \to \left[\frac{1}{n}\int \frac{[1-g(x)]^2}{(1-\varepsilon)+\varepsilon g(x)}\mathrm{d}F_0(x)\right]^{-\frac{1}{2}} = \frac{\tau_n^\star}{\sqrt{n}} \quad \Longrightarrow \quad \mathcal{V} = \frac{\tau_n^\star + o_g(1)}{\sqrt{n}}$$

where $o_g(1)$ denote a quantity that converges to zero when Δ_g converges to zero.

We then analyze the bias term $\mathcal{B}$. The numerator of the term $\mathcal{B}$ is bounded by

$$\begin{aligned}\left|\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x)\right]\right| &= \left|\int \frac{(1-\widehat{g}(x))(\widehat{g}(x)-g(x))}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\mathrm{d}F_0(x)\right| \\ &\le \Delta_g \cdot \int |1-\widehat{g}(x)|\mathrm{d}F_0(x) = 2\Delta_g \cdot D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}).\end{aligned}$$

For the denominator, by (19), we have

$$\begin{aligned}\left|\int \widehat{v}_{\mathrm{opt}}(\varepsilon, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]\right| &= \int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\mathrm{d}F_0(x) \\ &\ge \frac{D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0)}{C_g} \ge \frac{4}{C_g}D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}}).\end{aligned}$$

Here, $C_g \ge 1$ represents the upper bound of $|\widehat{g}(x)|$. Since $\sup_x |g(x)| < \infty$, we have $C_g < \infty$ as $\Delta_g \to 0$. This ensures that the bounds involving C_g remain finite under the given assumptions. Combining these bounds, the bias term is bounded by

$$\mathcal{B} \le \frac{C_g \Delta_g}{2D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}})} = o_g(1).$$

The last equation uses the fact that as $\Delta_g \to 0$, the total variation distance $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) \to D_{\mathrm{TV}}(F_0, \bar{F}_{\boldsymbol{P}}) > 0$.

Now suppose the moment condition $\mathbb{E}_{F_0}[g]^{-2} < \infty$ holds. This implies that $\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[g]^{-2} = \mathbb{E}_{F_0}[g]^{-1} \leq \sqrt{\mathbb{E}_{F_0}[g]^{-2}} < \infty$, and therefore $\mathbb{E}_{\bar{F}}[g]^{-2} \leq \max\{\mathbb{E}_{\bar{F}_{\boldsymbol{P}}}[g]^{-2}, \mathbb{E}_{F_0}[g]^{-2}\} < \infty$. When $\Delta_g \to 0$, we have $g(x) \leq (1+\Delta_g)\widehat{g}(x)$ for all x, which implies $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} \leq (1+\Delta_g)^2\,\mathbb{E}_{\bar{F}}[g]^{-2} < \infty$. This confirms that the moment condition in Corollary 5 holds. If $\varepsilon_{\min}$ is set as a small constant, then

$$A = \mathcal{O}\left(\Delta_g + \sqrt{\frac{\log n}{n}}\right) = o_g(1) + o(1).$$

If we set $\varepsilon_{\min} = \log n/\sqrt{n}$, then we obtain

$$A = \mathcal{O}\left(\Delta_g + \frac{1}{\log n} + \frac{1}{\sqrt{\log n}}\right) = o_g(1) + o(1).$$

Without the moment condition $\mathbb{E}_{F_0}[g]^{-2} < \infty$. Finally, if the moment condition $\mathbb{E}_{F_0}[g]^{-2} < \infty$ does not hold, then we only have

$$A = \mathcal{O}\left(\frac{\Delta_g}{\varepsilon_{\min}} + \frac{1}{\sqrt{\log^3 n}}\right) = \mathcal{O}\left(n^{1/4}\cdot\Delta_g\right) + o(1),$$

when we set $\varepsilon_{\min} = \frac{\log n}{n^{1/4}}$. To ensure that the above $A = o(1)$, we must impose the stronger requirement $n^{1/4}\cdot\Delta_g = o(1)$, which means the estimation error in $\widehat{g}$ must be significantly smaller than in the previous two cases. □

A.6.3 Proof of Lemma A.1

Proof of Lemma A.1. In the following, all integrations are taken with respect to x. For simplicity, we omit explicit notation for the dependence on x. By the definition of $\widehat{v}_{\text{opt}}(\varepsilon, x)$, and noting that $\widehat{g}(x) = \frac{\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}$, it follows that for any $\varepsilon \in [0,1]$,

$$\widetilde{\mathcal{T}}(\varepsilon) = \frac{\int \widehat{v}_{\text{opt}}(\varepsilon,x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}(x)\right]}{\int \widehat{v}_{\text{opt}}(\varepsilon,x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]} = \frac{\int \frac{1-\widehat{g}}{(1-\varepsilon)+\varepsilon\widehat{g}}\left[\mathrm{d}F_0 - \mathrm{d}\widehat{F}\right]}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0}.$$

Direct calculation yields that

$$\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} = \frac{\int \frac{[1-\widehat{g}]^2}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\left[\mathrm{d}F_0 - \mathrm{d}\widehat{F}\right]}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} - \frac{\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2}\cdot\int \frac{1-\widehat{g}}{(1-\varepsilon)+\varepsilon\widehat{g}}\left[\mathrm{d}F_0 - \mathrm{d}\widehat{F}\right].$$

Taking the expectation (with randomness only from $\widehat{F}$), we have

$$\mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} = \frac{\int \frac{[1-\widehat{g}]^2}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\left[\mathrm{d}F_0 - \mathrm{d}\bar{F}\right]}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} - \frac{\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2}\cdot\int \frac{1-\widehat{g}}{(1-\varepsilon)+\varepsilon\widehat{g}}\left[\mathrm{d}F_0 - \mathrm{d}\bar{F}\right].$$

To distinguish, we use ε_0 to denote the true proportion that generates $\widehat{F}$. Observe the equation that

$$\mathrm{d}\bar{F} = (1-\varepsilon_0)\mathrm{d}F_0 + \varepsilon_0\mathrm{d}\bar{F}_{\boldsymbol{P}} = [(1-\varepsilon_0) + \varepsilon_0 g]\,\mathrm{d}F_0.$$

Rearranging terms yields that

$$\begin{aligned}
\mathbb{E}\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} &= \varepsilon_0 \frac{\int \frac{[1-\widehat{g}]^2[1-g]}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} - \varepsilon_0 \frac{\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \cdot \int \frac{[1-\widehat{g}][1-g]}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0 \\
&= \varepsilon_0 \left[\frac{\int \frac{[1-\widehat{g}]^2[1-g]}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} - \frac{\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0}\right] \\
&\quad + \varepsilon_0 \frac{\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \cdot \left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0 - \int \frac{[1-\widehat{g}][1-g]}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right] \\
&= \frac{\varepsilon_0 \int \frac{[1-\widehat{g}]^2[\widehat{g}-g]}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} + \frac{\varepsilon_0 \int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \int \frac{[1-\widehat{g}][g-\widehat{g}]}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0. \qquad (16)
\end{aligned}$$

To proceed with the proof, we first derive individual bounds for each term in (16). Let $C_g = \max\{1, \sup_{x\in[0,1]} |\widehat{g}(x)|\}$. We also recall the definition of the total variation (TV) distance:

$$D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) := \frac{1}{2}\int |1-\widehat{g}(x)|\mathrm{d}F_0(x). \qquad (17)$$

Since $0 \le \widehat{g}(x) < \infty$ and $\varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]$, the following uniform bound holds:

$$-\frac{1}{\varepsilon_{\min}} \le -\frac{1}{\varepsilon} \le \frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)} \le \frac{1}{1-\varepsilon} \le \frac{1}{\varepsilon_{\min}}. \qquad (18)$$

(a) Note that $\frac{|\widehat{g}(x)-g(x)|}{\min\{1,\widehat{g}(x)\}} \le \Delta_g$ for all $x \in [0,1]$. As a result, we have that $|\widehat{g}(x)-g(x)| \le \Delta_g \cdot \min\{1, \widehat{g}(x)\} \le \Delta_g[(1-\varepsilon)+\varepsilon\widehat{g}(x)]$ for any $\varepsilon, x \in [0,1]$. Hence, the first term of (16) $\le \varepsilon_0\Delta_g$.

(b) By the bound in (18) and the definition of TV distance in (17), it follows that

$$\int \frac{|1-\widehat{g}(x)|^3}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2}\mathrm{d}F_0(x) \le \frac{1}{\varepsilon_{\min}} \cdot \int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\mathrm{d}F_0(x).$$

On the other hand, it follows that

$$\begin{aligned}
\int \frac{[1-\widehat{g}(x)]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2}\mathrm{d}F_0(x) &\le \int_{x:\widehat{g}(x)\le 1} \frac{[1-\widehat{g}(x)]^3}{\min\{1,\widehat{g}(x)\}^2}\mathrm{d}F_0(x) \\
&= \int_{x:\widehat{g}(x)\le 1} \frac{1}{[\widehat{g}(x)]^2}\mathrm{d}F_0(x) \le \int \frac{1}{[\widehat{g}(x)]^2}\mathrm{d}F_0(x).
\end{aligned}$$

For the other direction, it follows that

$$\int \frac{[\widehat{g}(x)-1]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2}\mathrm{d}F_0(x) \le \int_{x:\widehat{g}(x)\ge 1} \frac{[\widehat{g}(x)-1]^3}{\min\{1,\widehat{g}(x)\}^2}\mathrm{d}F_0(x)$$

$$= \int_{x:\widehat{g}(x)\geq 1} [\widehat{g}(x)]^3 \mathrm{d}F_0(x) \leq \int [\widehat{g}(x)]^3 \mathrm{d}F_0(x).$$

Combining the above bounds, we have

$$\left| \int \frac{[1-\widehat{g}(x)]^3 \mathrm{d}F_0(x)}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2} \right| \leq \min\left\{ \frac{1}{\varepsilon_{\min}} \int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)} \mathrm{d}F_0(x), \mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2} \right\}.$$

(*c*) By the definition of C_g, we have that

$$\int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)} \mathrm{d}F_0(x) \geq \frac{1}{C_g} \int [1-\widehat{g}(x)]^2 \mathrm{d}F_0(x) = \frac{D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0)}{C_g} \tag{19}$$

where $D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0) := \int [1-\widehat{g}(x)]^2 \mathrm{d}F_0(x)$ is the χ^2-divergence between $\widehat{F}_{\boldsymbol{P}}$ and F_0.

(*d*) The definition of $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}})$ implies that

$$\left| \int \frac{[1-\widehat{g}(x)][g(x)-\widehat{g}(x)]}{(1-\varepsilon)+\varepsilon\widehat{g}(x)} \mathrm{d}F_0(x) \right| \leq \Delta_g \cdot \int |1-\widehat{g}(x)| \mathrm{d}F_0(x) = 2\Delta_g D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}).$$

To establish (15b), we combine the bounds on the above four individual terms and apply them to the expression in (16). Note that $4D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}}) \leq D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0)$, $C_g \geq 1$, and $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) \leq 1$. We have that

$$\begin{aligned}
\left| \mathbb{E} \frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right| &\leq \varepsilon_0 \Delta_g \left[1 + \frac{C_g^2}{D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0)} \min\left\{ \frac{1}{\varepsilon_{\min}}, \frac{\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}}{\sqrt{D_{\chi^2}(\widehat{F}_{\boldsymbol{P}}, F_0)}} \right\} \right] \\
&\leq \varepsilon_0 \Delta_g \left[1 + \frac{C_g^2}{4D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}})} \min\left\{ \frac{1}{\varepsilon_{\min}}, \frac{\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}}{2D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}})} \right\} \right] \\
&\leq \varepsilon_0 \Delta_g \left[1 + \frac{C_g^2}{4D_{\mathrm{TV}}^3(F_0, \widehat{F}_{\boldsymbol{P}})} \min\left\{ \frac{1}{\varepsilon_{\min}}, \mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2} \right\} \right].
\end{aligned}$$

Lemma A.2 (High probability concentration)**.** *Define two classes of functions as*

$$\begin{aligned}
\mathcal{H}_1 &= \left\{ \frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)} : \varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}] \right\}, \\
\mathcal{H}_2 &= \left\{ \frac{[1-\widehat{g}(x)]^2}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2} : \varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}] \right\}.
\end{aligned} \tag{20}$$

There exists some universal constant $K > 0$ so that for every $C > 0$,

$$\begin{aligned}
\mathbb{P}\left(\sup_{h\in\mathcal{H}_1} \left| \frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)] \right| \geq \frac{1}{\varepsilon_{\min}} \sqrt{\frac{C\log n}{2n}} \right) &\leq \frac{4K}{\varepsilon_{\min}^3} \cdot \frac{1}{n^{C-1}}, \\
\mathbb{P}\left(\sup_{h\in\mathcal{H}_2} \left| \frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)] \right| \geq \frac{1}{\varepsilon_{\min}^2} \sqrt{\frac{C\log n}{2n}} \right) &\leq \frac{4K}{\varepsilon_{\min}^3} \cdot \frac{1}{n^{C-1}}.
\end{aligned} \tag{21}$$

If we further have $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$*, we then define* $M := 8\,\mathbb{E}_{\bar{F}}\left[\widehat{g} - 1/\widehat{g}\right]^2$*. There exists some universal constant* $K > 0$ *so that for every* $C > 0$*,*

$$\begin{aligned}
\mathbb{P}\left(\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C\log n}{n\varepsilon_{\min}} + \sqrt{\frac{CM\log n}{n}}\right) &\leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}},\\
\mathbb{P}\left(\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}\right) &\leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.
\end{aligned} \tag{22}$$

We now proceed to prove (15a). To this end, we utilize the high-probability concentration result established in Lemma A.2, whose proof is provided in Appendix A.7. We further consider two cases, depending on whether the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ holds.

Without the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$**.** Similar to the way we bounded the individual terms in (16), we have:

$$\frac{\int \frac{|1-\widehat{g}|^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \overset{(18)}{\leq} \frac{1}{\varepsilon_{\min}}\cdot\frac{1}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} \leq C_1, \quad \text{and} \quad \frac{1}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} \leq C_2,$$

where

$$C_1 := \frac{C_g}{4\varepsilon_{\min} D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}})} = \frac{C_2}{\varepsilon_{\min}}, \quad \text{and} \quad C_2 := \frac{C_g}{4D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}})}.$$

With the above inequalities in place, we are now ready to prove (15a):

$$\begin{aligned}
&\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]}\left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \geq \left(\frac{C_1}{\varepsilon_{\min}} + \frac{C_2}{\varepsilon_{\min}^2}\right)\sqrt{\frac{C\log n}{n}}\right)\\
&\leq \mathbb{P}\left(\frac{\int \frac{|1-\widehat{g}|^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2}\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C_1}{\varepsilon_{\min}}\sqrt{\frac{C\log n}{n}}\right)\\
&\quad + \mathbb{P}\left(\frac{1}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0}\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C_2}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{n}}\right)\\
&\leq \mathbb{P}\left(\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{1}{\varepsilon_{\min}}\sqrt{\frac{C\log n}{n}}\right)\\
&\quad + \mathbb{P}\left(\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{1}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{n}}\right)\\
&\overset{(21)}{\leq} \frac{8K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.
\end{aligned}$$

We can then rearrange the last inequality as follows:

$$\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]}\left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \geq \frac{C_g}{2\varepsilon_{\min}^2 D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}})}\sqrt{\frac{C\log n}{n}}\right) \leq \frac{8K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

We complete the proof by noting that $C_g \geq 1$ and $D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) \leq 1$.

With the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$. When the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ holds, it also implies that $\mathbb{E}_{F_0}[\widehat{g}]^{-2} < \infty$. It then follows that:

$$\frac{\left|\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0\right|}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \leq \frac{\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2} \leq C_1', \quad \text{and} \quad \frac{1}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0} \leq C_2,$$

where the constants C_1' and C_2 are defined as

$$C_1' = \frac{C_g^2\left[\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}\right]}{16D_{\mathrm{TV}}^4(F_0, \widehat{F}_{\boldsymbol{P}})}, \quad \text{and} \quad C_2 = \frac{C_g}{4D_{\mathrm{TV}}^2(F_0, \widehat{F}_{\boldsymbol{P}})}.$$

With the concentration result in (22) in mind, we define the following deviation thresholds:

$$t_1 = \frac{C\log n}{n\varepsilon_{\min}} + \sqrt{\frac{CM\log n}{n}}, \quad \text{and} \quad t_2 = \frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}.$$

With the above bounds and notation in place, we are now ready to prove (15a):

$$\begin{aligned}
&\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]}\left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \geq C_1't_1 + C_2t_2\right) \\
&\leq \mathbb{P}\left(\frac{\left|\int \frac{[1-\widehat{g}]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}]^2}\mathrm{d}F_0\right|}{\left[\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0\right]^2}\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq C_1't_1\right) \\
&\qquad + \mathbb{P}\left(\frac{1}{\int \frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}}\mathrm{d}F_0}\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq C_2t_2\right) \\
&\leq \mathbb{P}\left(\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq t_1\right) + \mathbb{P}\left(\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^n h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq t_2\right) \\
&\overset{(22)}{\leq} \frac{8K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.
\end{aligned}$$

Using the fact that $t_1 \leq t_2$ and defining $C_0' := C_1' + C_2$, we can now rearrange the last inequality as follows:

$$\mathbb{P}\left(\sup_{\varepsilon\in[\varepsilon_{\min},1-\varepsilon_{\min}]}\left|\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} - \mathbb{E}\,\frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon}\right| \geq C_0'\left[\frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}\right]\right) \leq \frac{8K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

For notation simplicity, noting that $C_g \geq 1, D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}}) \leq 1$, and $\mathbb{E}_{F_0}[\widehat{g}]^{-2} \geq [\mathbb{E}_{F_0}\,\widehat{g}]^{-2} = 1$, we introduce a slightly larger but more convenient upper bound C_0' given by

$$C_0' = \frac{C_g^2\left[\mathbb{E}_{F_0}[\widehat{g}]^3 + \mathbb{E}_{F_0}[\widehat{g}]^{-2}\right]}{D_{\mathrm{TV}}^4(F_0, \widehat{F}_{\boldsymbol{P}})}.$$

It is straightforward to verify that $C_0' \geq C_1' + C_2$ still holds. □

A.6.4 Proof of Theorem A.2

Proof of Theorem A.2. By the moment conditions, $0 < D_{\mathrm{TV}}(F_0, \widehat{F}_{\boldsymbol{P}})$. By definition, $\widehat{\varepsilon}_{\mathrm{opt}}$ is the fixed point of the equation $\varepsilon = \widehat{\mathcal{T}}(\varepsilon) = \mathcal{P} \circ \widetilde{\mathcal{T}}(\varepsilon)$, where the operators $\widehat{\mathcal{T}}, \mathcal{P}$ and $\widetilde{\mathcal{T}}$ are defined in (14). Let the true proportion that generates the observed samples $Y_1, \ldots, Y_n$ be denoted by ε_0. Then, ε_0 satisfies the fixed-point equation $\varepsilon_0 = \mathcal{T}(\varepsilon_0)$, where $\mathcal{T}$ is essentially an identity mapping:

$$\mathcal{T}(\varepsilon) = \frac{\int v_{\mathrm{opt}}(\varepsilon, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\bar{F}(x)\right]}{\int v_{\mathrm{opt}}(\varepsilon, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x)\right]},$$

and $v_{\mathrm{opt}}(\varepsilon, x)$ is the optimal weight function. To see this, we substitute the definition of v_{opt} and get

$$\mathcal{T}(\varepsilon) = \frac{\int \frac{1-g(x)}{(1-\varepsilon)+\varepsilon g(x)} \left[\mathrm{d}F_0(x) - \mathrm{d}\bar{F}(x)\right]}{\int \frac{[1-g(x)]^2}{(1-\varepsilon)+\varepsilon g(x)} \mathrm{d}F_0(x)} = \varepsilon.$$

As $\widehat{\varepsilon}_{\mathrm{opt}} \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]$ and $\varepsilon_0 \in [0,1]$ are solutions to their respective fixed-point equations, we have

$$\begin{aligned}
|\widehat{\varepsilon}_{\mathrm{opt}} - \varepsilon_0| &\le |\widehat{\varepsilon}_{\mathrm{opt}} - \mathcal{P}(\varepsilon_0)| + |\varepsilon_0 - \mathcal{P}(\varepsilon_0)| \\
&\overset{(a)}{\le} |\widehat{\mathcal{T}}(\widehat{\varepsilon}_{\mathrm{opt}}) - \mathcal{T} \circ \mathcal{P}(\varepsilon_0)| + \varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\} \\
&\le |\widehat{\mathcal{T}}(\widehat{\varepsilon}_{\mathrm{opt}}) - \widehat{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0)| + |\widehat{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0) - \mathcal{T} \circ \mathcal{P}(\varepsilon_0)| + \varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\} \\
&\overset{(b)}{\le} |\widetilde{\mathcal{T}}(\widehat{\varepsilon}_{\mathrm{opt}}) - \widetilde{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0)| + |\widetilde{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0) - \mathcal{T} \circ \mathcal{P}(\varepsilon_0)| + \varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\} \\
&\overset{(c)}{\le} L \cdot |\widehat{\varepsilon}_{\mathrm{opt}} - \mathcal{P}(\varepsilon_0)| + |\widetilde{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0) - \mathcal{T} \circ \mathcal{P}(\varepsilon_0)| + \varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\},
\end{aligned}$$

where (a) uses the relation that $|\varepsilon_0 - \mathcal{P}(\varepsilon_0)| \le \varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\}$. This is because we have $|\varepsilon_0 - \mathcal{P}(\varepsilon_0)| = 0$ if ε_0 is within the interval $[\varepsilon_{\min}, 1-\varepsilon_{\min}]$; otherwise, $|\varepsilon_0 - \mathcal{P}(\varepsilon_0)| \le \varepsilon_{\min}$. Here (b) uses the relation $\mathcal{P} \circ \mathcal{T} \circ \mathcal{P} = \mathcal{T} \circ \mathcal{P}$, and (c) uses the fact that $\widetilde{\mathcal{T}}$ is L-Lipschitz continuous, where

$$L = \sup_{\varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]} \left| \frac{\mathrm{d}\widetilde{\mathcal{T}}(\varepsilon)}{\mathrm{d}\varepsilon} \right|.$$

Rearranging the resulting inequality yields

$$|\widehat{\varepsilon}_{\mathrm{opt}} - \varepsilon_0| \le \frac{|\widetilde{\mathcal{T}} \circ \mathcal{P}(\varepsilon_0) - \mathcal{T} \circ \mathcal{P}(\varepsilon_0)| + 2\varepsilon_{\min} \mathbf{1}\{\varepsilon \notin [\varepsilon_{\min}, 1-\varepsilon_{\min}]\}}{1-L}. \tag{23}$$

Without loss of generality, we assume $\varepsilon_0 \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]$. If ε_0 lies outside this interval, we instead consider $\mathcal{P}(\varepsilon_0)$. Now, let us derive a bound for $\mathbb{E}\,|\widetilde{\mathcal{T}}(\varepsilon_0) - \mathcal{T}(\varepsilon_0)|$. It follows that

$$\begin{aligned}
&\mathbb{E}|\widetilde{\mathcal{T}}(\varepsilon_0) - \mathcal{T}(\varepsilon_0)| \\
&\quad = \mathbb{E}\,|\widetilde{\mathcal{T}}(\varepsilon_0) - \mathbb{E}\,\widetilde{\mathcal{T}}(\varepsilon_0) + \mathbb{E}\,\widetilde{\mathcal{T}}(\varepsilon_0) - \mathcal{T}(\varepsilon_0)| \\
&\quad \le \mathbb{E}\,|\widetilde{\mathcal{T}}(\varepsilon_0) - \mathbb{E}\,\widetilde{\mathcal{T}}(\varepsilon_0)| + |\,\mathbb{E}\,\widetilde{\mathcal{T}}(\varepsilon_0) - \mathcal{T}(\varepsilon_0)| \\
&\quad \le \sqrt{\mathrm{Var}(\widetilde{\mathcal{T}}(\varepsilon_0))} + \left| \frac{\int \widehat{v}_{\mathrm{opt}}(\varepsilon_0, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\bar{F}(x)\right]}{\int \widehat{v}_{\mathrm{opt}}(\varepsilon_0, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]} - \varepsilon_0 \right|
\end{aligned}$$

$$= \frac{\sqrt{\frac{1}{n}\int \left[\frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\right]^2 \mathrm{d}\bar{F}(x)}}{\int \frac{[1-\widehat{g}(x)]^2}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\mathrm{d}F_0(x)} + \varepsilon_0 \left| \frac{\int \widehat{v}_{\mathrm{opt}}(\varepsilon_0, x)\left[\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x) - \mathrm{d}\bar{F}_{\boldsymbol{P}}(x)\right]}{\int \widehat{v}_{\mathrm{opt}}(\varepsilon_0, x)\left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)\right]} \right|$$
$$=: \mathcal{V} + \mathcal{B}. \tag{24}$$

Finally, we analyze the Lipschitz constant L. By Lemma A.1, we have

$$\mathbb{P}(L \geq A) \leq \frac{8K}{\varepsilon_{\min}^3} \frac{1}{n^{C-1}}. \tag{25}$$

We then complete the proof by combining (23), (24) and (25). □

A.7 Proof of Lemma A.2

Proof of Lemma A.2. We use a uniform bound in conjunction with either Hoeffding's inequality or Bernstein's inequality to establish high-probability concentration results.

Lemma A.3 (Example 19.7 in [31])**.** *Let $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$ be a collection of measurable functions indexed by a bounded subset $\Theta \subset \mathbb{R}^d$. Suppose there exists a measurable function m such that*

$$|h_{\theta_1}(x) - h_{\theta_2}(x)| \leq m(x) \cdot \|\theta_1 - \theta_2\|, \quad \textit{for every } \theta_1, \theta_2.$$

If $\mathbb{E}_F[|m|] < \infty$, then there exists a constant K depending only on Θ and d, such that the bracketing numbers satisfy

$$N_{[]}(\varepsilon\, \mathbb{E}_F[|m|], \mathcal{H}, L_1(\bar{F})) \leq K \left(\frac{\operatorname{diam}\Theta}{\varepsilon}\right)^d, \quad \textit{for every } 0 < \varepsilon < \operatorname{diam}\Theta.$$

Here, $N_{[]}(\varepsilon, \mathcal{H}, L_1(F))$ is the bracketing number, which is the minimum number of ε-brackets required to cover $\mathcal{H}$ under $L_1(F)$.

We will apply Lemma A.3 to bound the bracketing number of the function classes $\mathcal{H}_1$ and $\mathcal{H}_2$ defined in (20). For these classes, the proportion ε serves as the parameter θ with domain $\Theta = [\varepsilon_{\min}, 1 - \varepsilon_{\min}]$ and dimensionality $d = 1$. The data distribution F is the mixture $\bar{F} = (1 - \varepsilon_0)F_0 + \varepsilon_0 \bar{F}_{\boldsymbol{P}}$, where ε_0 is the true proportion generating the observed samples. We define

$$m(x) := \sup_{x \in \mathcal{X}} \left| \frac{\partial h_\theta(x)}{\partial \theta} \right|,$$

which bounds the partial derivative of $h_\theta(x)$ with respect to θ. For $\mathcal{H}_1$ and $\mathcal{H}_2$, we explicitly compute:

$$\text{For } \mathcal{H}_1, \quad \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\left(\frac{1-\widehat{g}(x)}{(1-\varepsilon)+\varepsilon\widehat{g}(x)}\right) = \frac{(-1+\widehat{g}(x))^2}{(1-\varepsilon+\varepsilon\widehat{g}(x))^2},$$

$$\text{For } \mathcal{H}_2, \quad \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\left(\frac{[1-\widehat{g}(x)]^2}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^2}\right) = \frac{2[1-\widehat{g}(x)]^3}{[(1-\varepsilon)+\varepsilon\widehat{g}(x)]^3}.$$

With $\varepsilon \in [\varepsilon_{\min}, 1 - \varepsilon_{\min}]$ and (18), we set $m(x) \equiv \frac{2}{\varepsilon_{\min}^3}$ as a uniform upper bound. Using Lemma A.3, we conclude that:

$$N_{[]}(\delta, \mathcal{H}_1, L_1(\bar{F})) \vee N_{[]}(\delta, \mathcal{H}_2, L_1(\bar{F})) \leq \frac{2K}{\varepsilon_{\min}^3} \cdot \frac{1}{\delta}.$$

Next, consider $\mathcal{H}_k$ for $k = 1, 2$. Define a finite set $\mathcal{H}_{\text{bracket}} = \{h_1, h_2, \ldots, h_{n'}\}$ as a $\frac{1}{n}$-$L_1(\bar{F})$-bracket of $\mathcal{H}_k$. By definition, for any $h \in \mathcal{H}_k$, there exists $h_i, h_j \in \mathcal{H}_{\text{bracket}}$ such that $h_i \leq h \leq h_j$ and $|\mathbb{E}_{\bar{F}}[h_i - h_j]| \leq \frac{1}{n}$. Using the bound on bracketing numbers, we have $n' \leq C_1 n$. Then:

$$\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)] \leq \frac{1}{n}\sum_{t=1}^{n} h_j(Y_t) - \mathbb{E}_{\bar{F}}[h_i(Y)] \leq \frac{1}{n}\sum_{t=1}^{n} h_j(Y_t) - \mathbb{E}_{\bar{F}}[h_j(Y)] + \frac{1}{n},$$

$$\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)] \geq \frac{1}{n}\sum_{t=1}^{n} h_i(Y_t) - \mathbb{E}_{\bar{F}}[h_j(Y)] \geq \frac{1}{n}\sum_{t=1}^{n} h_i(Y_t) - \mathbb{E}_{\bar{F}}[h_i(Y)] - \frac{1}{n}.$$

This implies that for any $t \geq \frac{1}{n}$,

$$\left\{\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| > t\right\} \subseteq \left\{\exists h_j \in \mathcal{H}_{\text{bracket}},\ \left|\frac{1}{n}\sum_{t=1}^{n} h_j(Y_t) - \mathbb{E}_{\bar{F}}[h_j(Y)]\right| > t - \frac{1}{n}\right\}.$$

Thus,

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| > t\right) \leq \sum_{h\in\mathcal{H}_{\text{bracket}}} \mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| > t - \frac{1}{n}\right).$$

A.7.1 With Moment Conditions

We first consider the case where the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ holds. Recall that $\sup_x |\widehat{g}(x)| < \infty$. Under these conditions, any function in either $\mathcal{H}_1$ or $\mathcal{H}_2$ has bounded variance, which is independent of—or at most weakly dependent on $\frac{1}{\varepsilon_{\min}}$. We begin by analyzing the variance for functions in $\mathcal{H}_1$. To this end, we consider two cases:

1. For $\varepsilon \in [\varepsilon_{\min}, \frac{1}{2}]$:

$$\text{Var}_{\bar{F}}\left(\frac{1-\widehat{g}}{[1-\varepsilon)+\varepsilon\widehat{g}}\right) \leq \mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}]^2}\right) \leq \frac{\mathbb{E}_{\bar{F}}\,[1-\widehat{g}]^2}{(1-\varepsilon)^2} \leq 4\,\mathbb{E}_{\bar{F}}\,[1-\widehat{g}]^2 < \infty.$$

2. For $\varepsilon \in [\frac{1}{2}, 1-\varepsilon_{\min}]$:

$$\text{Var}_{\bar{F}}\left(\frac{1-\widehat{g}}{(1-\varepsilon)+\varepsilon\widehat{g}}\right) \leq \mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}(x)]^2}{[\varepsilon\widehat{g}(x)]^2}\right) \leq 4\,\mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}(x)]^2}{[\widehat{g}(x)]^2}\right) < \infty.$$

Thus, the variance is bounded, and we define

$$4\,\mathbb{E}_{\bar{F}}\,[1-\widehat{g}]^2 + \mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}(x)]^2}{[\widehat{g}(x)]^2}\right) \leq 4\,\mathbb{E}_{\bar{F}}\,[\widehat{g}-1/\widehat{g}]^2 =: \frac{M}{2}.$$

Similarly, for the variance of functions in $\mathcal{H}_2$, we have, for any $\varepsilon \in [\varepsilon\min, 1-\varepsilon_{\min}]$:

$$\text{Var}_{\bar{F}}\left(\frac{[1-\widehat{g}]^2}{[1-\varepsilon)+\varepsilon\widehat{g}]^2}\right) \leq \mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}]^4}{(1-\varepsilon)+\varepsilon\widehat{g}]^4}\right) \overset{(18)}{\leq} \frac{1}{\varepsilon_{\min}^2}\,\mathbb{E}_{\bar{F}}\left(\frac{[1-\widehat{g}]^2}{(1-\varepsilon)+\varepsilon\widehat{g}]^2}\right) \leq \frac{M}{2\varepsilon_{\min}^2}.$$

For $\mathcal{H}_1$. Using Bernstein's inequality, we have the following bound for any $h_1 \in \mathcal{H}_1$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^{n} h_1(Y_t) - \mathbb{E}_{\bar{F}}[h_1(Y)]\right| > t - \frac{1}{n}\right) \leq 2\exp\left(-\frac{n(t-\frac{1}{n})^2}{2\left(\frac{M}{2} + \frac{(t-\frac{1}{n})}{3n\varepsilon_{\min}}\right)}\right).$$

Let $t \geq \frac{1}{n} + \frac{C\log n}{n\varepsilon_{\min}} + \sqrt{\frac{CM\log n}{n}}$ with any $C > 0$. Then it follows that

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C\log n}{n\varepsilon_{\min}} + \sqrt{\frac{CM\log n}{n}}\right) \leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

For $\mathcal{H}_2$. Using Bernstein's inequality, we have the following bound for any $h_2 \in \mathcal{H}_2$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^{n} h_2(Y_t) - \mathbb{E}_{\bar{F}}[h_2(Y)]\right| > t - \frac{1}{n}\right) \leq 2\exp\left(-\frac{n(t-\frac{1}{n})^2}{2\left(\frac{M}{2\varepsilon_{\min}^2} + \frac{(t-\frac{1}{n})}{3n\varepsilon_{\min}^2}\right)}\right).$$

Let $t \geq \frac{1}{n} + \frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}$ with any $C > 0$. Then it follows that

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{C\log n}{n\varepsilon_{\min}^2} + \sqrt{\frac{CM\log n}{n\varepsilon_{\min}^2}}\right) \leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

A.7.2 Without Moment Conditions

Finally, if the moment condition $\mathbb{E}_{\bar{F}}[\widehat{g}]^{-2} < \infty$ does not hold, we instead apply Hoeffding's inequality to bound the probability on the right-hand side for both $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively. As a result, the resulting concentration bound will depend on $\frac{1}{\varepsilon_{\min}^2}$.

For $\mathcal{H}_1$. For each $h \in \mathcal{H}_1$, we have $\sup_x |h(x)| \leq \frac{1}{\varepsilon_{\min}}$ due to $\varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]$. Applying Hoeffding's inequality, we obtain:

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| > t - \frac{1}{n}\right) \leq 2\exp\left(-2n\varepsilon_{\min}^2\left(t-\frac{1}{n}\right)^2\right).$$

Let $t \geq \frac{1}{n} + \frac{1}{\varepsilon_{\min}}\sqrt{\frac{C\log n}{2n}}$ with any positive constant $C > 0$. Then,

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}_1}\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{1}{\varepsilon_{\min}}\sqrt{\frac{C\log n}{2n}}\right) \leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

For $\mathcal{H}_2$. For each $h \in \mathcal{H}_2$, we similarly have $\sup_x |h(x)| \leq \frac{1}{\varepsilon_{\min}^2}$ due to $\varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]$. Applying Hoeffding's inequality, we obtain:

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^{n} h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| > t - \frac{1}{n}\right) \leq 2\exp\left(-2n\varepsilon_{\min}^4\left(t-\frac{1}{n}\right)^2\right).$$

Let $t \geq \frac{1}{n} + \frac{1}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{2n}}$ with any positive constant $C > 0$. Then,

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}_2}\left|\frac{1}{n}\sum_{t=1}^{n}h(Y_t) - \mathbb{E}_{\bar{F}}[h(Y)]\right| \geq \frac{1}{\varepsilon_{\min}^2}\sqrt{\frac{C\log n}{2n}}\right) \leq \frac{4K}{\varepsilon_{\min}^3}\cdot\frac{1}{n^{C-1}}.$$

□

A.8 Proof of Theorem 4.4

Proof of Theorem 4.4. We will use Le Cam's method to prove the minimax lower bound [34, Chapter 15.2]. From Proposition 15.1 and (15.14) in [34], it follows that

$$\inf_{\widehat{\varepsilon}\in\mathcal{G}}\sup_{\varepsilon_0\in\mathcal{B}(\varepsilon)}\mathbb{E}\left|\widehat{\varepsilon} - \varepsilon_0\right| \geq \inf_{\widehat{\varepsilon}\in\mathcal{G}}\sup_{\varepsilon_0\in\{\varepsilon_1,\varepsilon_2\}}\mathbb{E}\left|\widehat{\varepsilon} - \varepsilon_0\right| \geq \frac{|\varepsilon_1 - \varepsilon_2|}{2}\left(1 - D_{\mathrm{TV}}(\rho_1,\rho_2)\right). \tag{26}$$

Here, ε_i for $i = 1, 2$ are carefully chosen proportions within $\mathcal{B}(\varepsilon)$, and ρ_i denotes the joint probability distribution of the considered functional of pivotal statistics in $\mathcal{G}$ when the underlying proportion is ε_i. Recall that each $Y_t|\boldsymbol{P}$ is generated according to $(1-\varepsilon_i)\mu_0 + \varepsilon_i\mu_{1,\boldsymbol{P}_t}$, where $\boldsymbol{P}_t$ is independently and uniformly sampled from the set $\{\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n\}$. We explain the choice of ε_i as follows. Since $\mathcal{B}(\varepsilon)$ is an open interval containing ε, we must have $\varepsilon \in \mathcal{B}(\varepsilon)$ and $\varepsilon + \Delta \in \mathcal{B}(\varepsilon)$ for a sufficiently small Δ. We then select $\varepsilon_1 = \varepsilon + \Delta$ and $\varepsilon_2 = \varepsilon$.

In the following, we discuss two classes of estimators and identify the minimax lower bound for their mean absolute error. The proof ideas are essentially the same.

(a) Indicator functions. Let $\mathcal{G}$ collect all measurable functions of the n-dimensional indicator functions $(\mathbf{1}\{Y_1 \leq \delta\}, \ldots, \mathbf{1}\{Y_n \leq \delta\})$. We first establish the lower bound for $\widehat{\varepsilon} \in \mathcal{G}$. Given that the Y_t's are generated independently and identically, the joint distribution ρ_i is the product of n independent Bernoulli distributions:

$$\rho_i = \mathrm{Ber}(p(\varepsilon_i))^{\otimes n},$$

where

$$p(\varepsilon_i) = (1-\varepsilon_i)F_0(\delta) + \varepsilon_i\bar{F}_{\boldsymbol{P}}(\delta).$$

By the relation (2) and the choice that $\varepsilon_2 = \varepsilon$, it follows that

$$\bar{F}(\delta) = (1-\varepsilon)F_0(\delta) + \varepsilon\bar{F}_{\boldsymbol{P}}(\delta) = p(\varepsilon) = p(\varepsilon_2). \tag{27}$$

Next, we further upper bound the total variation distance $D_{\mathrm{TV}}(\rho_1, \rho_2)$ in (26) to simplify the expression.

$$\begin{aligned} D_{\mathrm{TV}}(\rho_1,\rho_2) &\overset{(a)}{\leq} \sqrt{\frac{1}{2}D_{\mathrm{KL}}(\rho_1,\rho_2)} \\ &\overset{(b)}{\leq} \sqrt{\frac{n}{2}D_{\mathrm{KL}}(\mathrm{Ber}(p(\varepsilon_1)),\mathrm{Ber}(p(\varepsilon_2)))} \\ &\overset{(c)}{\leq} \sqrt{\frac{n|\varepsilon_1-\varepsilon_2|^2}{2}\frac{[F_0(\delta)-\bar{F}_{\boldsymbol{P}}(\delta)]^2}{p(\varepsilon_2)(1-p(\varepsilon_2))}} \end{aligned}$$

$$\overset{(d)}{\leq} \frac{1}{\sqrt{2}}|F_0(\delta) - \bar{F}_{\boldsymbol{P}}(\delta)|\Delta \cdot \sqrt{\frac{n}{\bar{F}(\delta)(1-\bar{F}(\delta))}}$$
$$\overset{(e)}{\leq} \sqrt{\frac{n}{2}}\frac{\Delta}{\sigma_n^\star}, \tag{28}$$

where

- (a) uses the Pinsker inequality, which allows us to upper bound the total variation distance by the square root of half the Kullback-Leibler (KL) divergence.
- (b) employs the decoupling property of KL divergences and the fact that $\rho_i = \mathrm{Ber}(p(\varepsilon_i))^{\otimes n}$, leading to
$$D_{\mathrm{KL}}(\rho_1 \| \rho_2) = n D_{\mathrm{KL}}(\mathrm{Ber}(p(\varepsilon_1)) \| \mathrm{Ber}(p(\varepsilon_2))).$$
- (c) utilizes the numerical inequality that
$$D_{\mathrm{KL}}(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) = p \log \frac{p}{q} + (1-p)\log\frac{1-p}{1-q} \leq \frac{(p-q)^2}{q(1-q)}.$$
- (d) uses the notation $\Delta = |\varepsilon_1 - \varepsilon_2|$.
- (e) uses the definition of $\sigma_n^\star$ that is $[\sigma_n^\star]^2 = \frac{\bar{F}(\varepsilon)(1-\bar{F}(\varepsilon))}{[F_0(\delta) - \bar{F}_{\boldsymbol{P}}(\delta)]^2}$.

Combining (26) and (28), we obtain

$$\inf_{\widehat{\varepsilon} \in \mathcal{G}} \sup_{\varepsilon_0 \in \mathcal{B}(\varepsilon)} \mathbb{E}\,|\widehat{\varepsilon} - \varepsilon_0| \geq \frac{\Delta}{2}\left(1 - \frac{1}{\sqrt{2}}\frac{\sqrt{n}\Delta}{\sigma_n^\star}\right).$$

By setting $\Delta = \frac{\sigma_n^\star}{\sqrt{2n}}$ and assuming that n is sufficiently large such that $\varepsilon + \Delta \in \mathcal{B}(\varepsilon)$, we obtain

$$\inf_{\widehat{\varepsilon} \in \mathcal{G}} \sup_{\varepsilon_0 \in \mathcal{B}(\varepsilon)} \mathbb{E}\,|\widehat{\varepsilon} - \varepsilon_0| \geq \frac{\sigma_n^\star}{\sqrt{2n}}.$$

This completes the proof of the first part.

(b) General functions. Let $\mathcal{G}$ denote the set of all general measurable functions of the n-dimensional random vector $(Y_1, \ldots, Y_n)$. We now establish the lower bound for any $\widehat{\varepsilon} \in \mathcal{G}$ under this broader class of functions. Given that the Y_t's are generated independently, the joint distribution ρ_i is the product of n independent mixture distributions:

$$\rho_i = [(1-\varepsilon_i)F_0 + \varepsilon_i \bar{F}_{\boldsymbol{P}}]^{\otimes n}.$$

We similarly begin with (26) and aim to upper bound the total variation distance $D_{\mathrm{TV}}(\rho_1, \rho_2)$. Since the proof closely resembles that of (28), we focus only on the key differences below.

$$D_{\mathrm{TV}}(\rho_1, \rho_2) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\rho_1, \rho_2)}$$
$$= \sqrt{\frac{n}{2} D_{\mathrm{KL}}((1-\varepsilon_1)F_0 + \varepsilon_1 \bar{F}_{\boldsymbol{P}}, (1-\varepsilon_2)F_0 + \varepsilon_2 \bar{F}_{\boldsymbol{P}})}$$

$$
\begin{aligned}
&\overset{(a)}{\leq} \sqrt{\frac{n}{2} D_{\chi^2}((1-\varepsilon_1)F_0 + \varepsilon_1 \bar{F}_{\boldsymbol{P}}, (1-\varepsilon_2)F_0 + \varepsilon_2 \bar{F}_{\boldsymbol{P}})} \\
&\overset{(b)}{\leq} \sqrt{\frac{|\varepsilon_1 - \varepsilon_2|^2}{2} n \cdot \int \frac{(\mathrm{d}F_0 - \mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0 + \varepsilon \mathrm{d}\bar{F}_{\boldsymbol{P}}}} \\
&= \Delta \sqrt{\frac{n}{2}} \cdot \sqrt{\int \frac{(\mathrm{d}F_0 - \mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0 + \varepsilon \mathrm{d}\bar{F}_{\boldsymbol{P}}}} \\
&\overset{(c)}{=} \sqrt{\frac{n}{2}} \frac{\Delta}{\tau_n^\star},
\end{aligned} \tag{29}
$$

where

(a) uses the fact that the Kullback-Leibler (KL) divergence is always upper bounded by its Chi-square divergence:

$$
D_{\mathrm{KL}}(P, Q) = \int \log \frac{\mathrm{d}P}{\mathrm{d}Q} \mathrm{d}P \leq \int \frac{(\mathrm{d}P - \mathrm{d}Q)^2}{\mathrm{d}Q} = D_{\chi^2}(P, Q).
$$

(b) uses the fact that

$$
D_{\chi^2}((1-\varepsilon_1)F_0 + \varepsilon_1 \bar{F}_{\boldsymbol{P}}, (1-\varepsilon_2)F_0 + \varepsilon_2 \bar{F}_{\boldsymbol{P}}) = |\varepsilon_1 - \varepsilon_2|^2 \cdot \int \frac{(\mathrm{d}F_0 - \mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0 + \varepsilon \mathrm{d}\bar{F}_{\boldsymbol{P}}} \tag{30}
$$

(c) uses the definition of $\tau_n^\star$,, which is given by:

$$
[\tau_n^\star]^2 = \left[\int \frac{(\mathrm{d}F_0 - \mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0 + \varepsilon \mathrm{d}\bar{F}_{\boldsymbol{P}}} \right]^{-1} = \left[\int \frac{(1-g(x))^2}{(1-\varepsilon) + \varepsilon g(x)} \mathrm{d}F_0(x) \right]^{-1} \quad \text{where} \quad g(x) = \frac{\mathrm{d}\bar{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}.
$$

The remaining argument follows the same reasoning as in part (a). In this way, we will have

$$
\inf_{\widehat{\varepsilon} \in \mathcal{G}} \sup_{\varepsilon_0 \in \mathcal{B}(\varepsilon)} \mathbb{E} \, |\widehat{\varepsilon} - \varepsilon_0| \geq \frac{\tau_n^\star}{\sqrt{2n}}.
$$

(c) Final inequality. In the end, we prove the inequality that $\sigma_n^\star \geq \tau_n^\star$. The data processing inequality [30] states that any post-processing of two distributions reduces the value of the f-divergence between them, including the Chi-square divergence. Therefore, using the same choices of ε_1, ε_2 as before, we have:

$$
\begin{aligned}
&D_{\chi^2}(\mathrm{Ber}(p(\varepsilon_1)), \mathrm{Ber}(p(\varepsilon_2))) \\
&\quad = D_{\chi^2}(\mathrm{Ber}((1-\varepsilon_1)F_0(\delta) + \varepsilon_1 \bar{F}_{\boldsymbol{P}}(\delta)), \mathrm{Ber}((1-\varepsilon_2)F_0(\delta) + \varepsilon_2 \bar{F}_{\boldsymbol{P}}(\delta))) \\
&\quad \leq D_{\chi^2}((1-\varepsilon_1)F_0 + \varepsilon_1 \bar{F}_{\boldsymbol{P}}, (1-\varepsilon_2)F_0 + \varepsilon_2 \bar{F}_{\boldsymbol{P}}).
\end{aligned} \tag{31}
$$

Note that, by definition, it follows that:

$$
D_{\chi^2}(\mathrm{Ber}(p(\varepsilon_1)), \mathrm{Ber}(p(\varepsilon_2))) = \frac{[p(\varepsilon_1) - p(\varepsilon_2)]^2}{p(\varepsilon_2)(1-p(\varepsilon_2))} = |\varepsilon_1 - \varepsilon_2|^2 \cdot \frac{[F_0(\delta) - \bar{F}_{\boldsymbol{P}}(\delta)]^2}{\bar{F}(\delta)(1-\bar{F}(\delta))}, \tag{32}
$$

$$D_{\chi^2}((1-\varepsilon_1)F_0+\varepsilon_1\bar{F}_{\boldsymbol{P}},(1-\varepsilon_2)F_0+\varepsilon_2\bar{F}_{\boldsymbol{P}})=|\varepsilon_1-\varepsilon_2|^2\cdot\int\frac{(\mathrm{d}F_0-\mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0+\varepsilon\mathrm{d}\bar{F}_{\boldsymbol{P}}}. \tag{33}$$

By combining (31), (32), and (33), we have that

$$\frac{1}{[\tau_n^\star]^2}=\int\frac{(\mathrm{d}F_0-\mathrm{d}\bar{F}_{\boldsymbol{P}})^2}{(1-\varepsilon)\mathrm{d}F_0+\varepsilon\mathrm{d}\bar{F}_{\boldsymbol{P}}}\geq\frac{[F_0(\delta)-\bar{F}_{\boldsymbol{P}}(\delta)]^2}{\bar{F}(\delta)(1-\bar{F}(\delta))}=\frac{1}{[\sigma_n^\star]^2}\quad\Longrightarrow\quad\sigma_n^\star\geq\tau_n^\star.$$

□

A.9 Proof of Theorem 4.5

Proof of Theorem 4.5. For clarity, let $(\widehat{\varepsilon}_\lambda,\widehat{\mu}_\lambda)$ denote the solution to the L_2-regularized maximum likelihood estimation (MLE) with regularization parameter $\lambda>0$.

By definition, the L_2-regularized MLE is given by:

$$(\widehat{\varepsilon}_\lambda,\widehat{\mu}_\lambda)=\arg\min_{\varepsilon,\mu}\left\{-\log L(\varepsilon,\mu)+\lambda(\mu^2+\varepsilon^2)\right\},$$

where

$$L(\varepsilon,\mu)=[(1-\varepsilon)\gamma+\varepsilon\mu]^{\widehat{e}}\,[1-(1-\varepsilon)\gamma-\varepsilon\mu]^{1-\widehat{e}}.$$

The solution $(\widehat{\varepsilon}_\lambda,\widehat{\mu}_\lambda)$ has several key properties:

(a) **Existence and uniqueness**: The solution exists and is unique due to the strong convexity introduced by the L_2-regularization term.

(b) **Domain restriction**: The condition $(1-\widehat{\varepsilon}_\lambda)\gamma+\widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda\in(0,1)$ must hold; otherwise, the optimization problem is ill-defined.

(c) **First-order conditions**: The solution $(\widehat{\varepsilon}_\lambda,\widehat{\mu}_\lambda)$ satisfies the following first-order conditions:

$$2\lambda\varepsilon=(\mu-\gamma)\left[\frac{\widehat{e}}{(1-\varepsilon)\gamma+\varepsilon\mu}-\frac{1-\widehat{e}}{1-(1-\varepsilon)\gamma-\varepsilon\mu}\right], \tag{34a}$$

$$2\lambda\mu=\varepsilon\left[\frac{\widehat{e}}{(1-\varepsilon)\gamma+\varepsilon\mu}-\frac{1-\widehat{e}}{1-(1-\varepsilon)\gamma-\varepsilon\mu}\right]. \tag{34b}$$

(d) **Bound on the mixture**: We claim that $(1-\widehat{\varepsilon}_\lambda)\gamma+\widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda<\widehat{e}$. To see this, suppose the equality $(1-\widehat{\varepsilon}_\lambda)\gamma+\widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda=\widehat{e}$ holds. Then the first-order conditions imply $\widehat{\varepsilon}_\lambda=\widehat{\mu}_\lambda=0$, which would force $\widehat{e}=\gamma$. This is impossible when n is sufficiently large because $\widehat{e}$ converges almost surely to $(1-\varepsilon)\gamma+\varepsilon\mu>\gamma$, as $\mu>\gamma$ and $1>\varepsilon>0$. Furthermore, by (34b), the condition $(1-\widehat{\varepsilon}_\lambda)\gamma+\widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda>\widehat{e}$ would imply $\widehat{\mu}_\lambda\widehat{\varepsilon}_\lambda<0$, which is a contradiction.

(e) **Limits as $\lambda\to0$**: The following limits exist:

$$\widehat{\varepsilon}=\lim_{\lambda\to0}\widehat{\varepsilon}_\lambda,\quad\widehat{\mu}=\lim_{\lambda\to0}\widehat{\mu}_\lambda. \tag{35}$$

This will be proven at the end of the proof.

By dividing (34a) by (34b), we obtain

$$\frac{\widehat{\varepsilon}_\lambda}{\widehat{\mu}_\lambda} = \frac{\widehat{\mu}_\lambda - \gamma}{\widehat{\varepsilon}_\lambda},$$

which simplifies to $\widehat{\varepsilon}_\lambda^2 = \widehat{\mu}_\lambda(\widehat{\mu}_\lambda - \gamma)$. Taking the limit as $\lambda \to 0$ and using (35), we have

$$\widehat{\varepsilon}^2 = \widehat{\mu}(\widehat{\mu} - \gamma). \tag{36}$$

We now claim that in the limit $\lambda \to 0$,

$$(1 - \widehat{\varepsilon})\gamma + \widehat{\varepsilon}\widehat{\mu} = \widehat{e}. \tag{37}$$

This follows from the fact that $\widehat{e} = \arg\max_{x \in (0,1)} x^{\widehat{e}}(1-x)^{1-\widehat{e}}$. Thus, any (ε, μ) maximizing $L(\varepsilon, \mu)$ when $\lambda = 0$ must satisfy (37).

Finally, we argue that $\widehat{\varepsilon} > 0$ and $\widehat{\mu} > \gamma$ as long as $\varepsilon > 0$. If this were not true, then for sufficiently large n, the strong law of large numbers implies $\widehat{e} \overset{a.s.}{\to} (1-\varepsilon)\gamma + \varepsilon\mu$, which would contradict the inequality that $\widehat{e} > \gamma$ from (37) when $\widehat{\varepsilon} = 0$.

Combining (36) and (37), and letting $x = \sqrt{\widehat{\mu} - \gamma}$ (recall that $\widehat{\mu} > \gamma$ for $\varepsilon > 0$), we find:

$$\widehat{\varepsilon} = x\sqrt{x^2 + \gamma}, \quad \widehat{\mu} = x^2 + \gamma,$$

where $x > 0$ satisfies

$$x^3\sqrt{x^2 + \gamma} = \widehat{e} - \gamma.$$

□

Proof of the existence of limits in (35). Rearranging (34a), we obtain:

$$(\widehat{\mu}_\lambda - \gamma)[\widehat{e} - (1 - \widehat{\varepsilon}_\lambda)\gamma - \widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda] = 2\lambda\widehat{\varepsilon}_\lambda \cdot [(1 - \widehat{\varepsilon}_\lambda)\gamma + \widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda][1 - (1 - \widehat{\varepsilon}_\lambda)\gamma - \widehat{\varepsilon}_\lambda\widehat{\mu}_\lambda]. \tag{38}$$

Combining (36) and (38), and defining $x_\lambda = \sqrt{\widehat{\mu}_\lambda - \gamma}$, we have:

$$\widehat{\varepsilon}_\lambda = x_\lambda\sqrt{x_\lambda^2 + \gamma}, \quad \widehat{\mu}_\lambda = x_\lambda^2 + \gamma,$$

and

$$x_\lambda^5\sqrt{x_\lambda^2 + \gamma} \le (\widehat{e} - \gamma) \cdot x_\lambda^2.$$

The inequality above implies that $x_\lambda > 0$ is uniformly bounded for any λ. Consequently, as a function of x_λ, both $\widehat{\varepsilon}_\lambda$ and $\widehat{\mu}_\lambda$ are uniformly bounded for all λ. As $\lambda \to 0$, we can extract a subsequence for which $\widehat{\varepsilon}_\lambda$ and $\widehat{\mu}_\lambda$ converge to well-defined limits. For this particular subsequence, taking the limit $\lambda \to 0$ in (38) yields:

$$\lim_{\lambda \to 0} x_\lambda^5\sqrt{x_\lambda^2 + \gamma} - (\widehat{e} - \gamma) \cdot x_\lambda^2 = 0.$$

Since $x_\lambda > 0$, the above equation has only one nonzero solution. Therefore, the subsequential limit of x_λ is unique. Consequently, all subsequences of $\widehat{\varepsilon}_\lambda$ and $\widehat{\mu}_\lambda$ converge to the same limits.

□

B Simulation Study: Details and Results

B.1 Further Details

Evaluation of the fixed-point equation. Recall that we need to solve the following fixed-point equation

$$\varepsilon = \left[\frac{\int \widehat{v}_{\text{opt}}(\varepsilon, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}(x) \right]}{\int \widehat{v}_{\text{opt}}(\varepsilon, x) \left[\mathrm{d}F_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x) \right]} \right]_{[\varepsilon_{\min}, 1-\varepsilon_{\min}]} \tag{8}$$

where $\widehat{v}_{\text{opt}}$ is the estimated optimal weight function defined by:

$$\widehat{v}_{\text{opt}}(\varepsilon, x) = \frac{1 - \widehat{g}(x)}{(1-\varepsilon) + \varepsilon \widehat{g}(x)}, \quad \text{and} \quad \widehat{g}(x) = \frac{\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)}{\mathrm{d}F_0(x)}.$$

We use the notation $[a]_{[b,c]} = \max\{\min\{a, c\}, b\}$ to denote the projection of a onto the interval $[b, c]$, ensuring that the result lies within the bounds of b and c.

In our implementation, we first sample $n = 10^6$ values, denoted as $Z_1, \ldots, Z_n$, from F_0 (which is $U(0, 1)$ due to the probability integral transform) and approximate $\int \widehat{v}_{\text{opt}}(\varepsilon, x)\mathrm{d}F_0(x)$ using the average over these samples: $\frac{1}{n}\sum_{t=1}^{n} \widehat{v}_{\text{opt}}(\varepsilon, Z_t)$. We denote the empirical sum by $\int \widehat{v}_{\text{opt}}(\varepsilon, x)\mathrm{d}\widehat{F}_0(x)$. Similarly, we collect the same number of pivotal statistics from $\bar{F}$ or $\bar{F}_{\boldsymbol{P}}$ and compute $\int \widehat{v}_{\text{opt}}(\varepsilon, x)\mathrm{d}\widehat{F}(x)$ or $\int \widehat{v}_{\text{opt}}(\varepsilon, x)\mathrm{d}\widehat{F}_{\boldsymbol{P}}(x)$ as their respective averages.

Optimization for the refined estimator. To obtain the refined estimator $\widehat{\varepsilon}_{\text{opt}}$, we use numerical methods to solve the following minimization problem:

$$\widehat{\varepsilon}_{\text{opt}} = \arg \min_{\varepsilon \in [\varepsilon_{\min}, 1-\varepsilon_{\min}]} \mathcal{L}(\varepsilon),$$

where $\varepsilon_{\min} = 10^{-3}$ is a small constant to avoid numerical issue,

$$\mathcal{L}(\varepsilon) = \left| \varepsilon - \widehat{\mathcal{T}}_{\text{used}}(\varepsilon) \right| \quad \text{and} \quad \widehat{\mathcal{T}}_{\text{used}}(\varepsilon) = \left[\frac{\int \widehat{v}_{\text{opt}}(\varepsilon, x) \left[\mathrm{d}\widehat{F}_0(x) - \mathrm{d}\widehat{F}(x) \right]}{\int \widehat{v}_{\text{opt}}(\varepsilon, x) \left[\mathrm{d}\widehat{F}_0(x) - \mathrm{d}\widehat{F}_{\boldsymbol{P}}(x) \right]} \right]_{[\varepsilon_{\min}, 1-\varepsilon_{\min}]}. \tag{39}$$

Here, $\mathcal{L}(\varepsilon)$ represents the deviation between ε and the estimated operator $\widehat{\mathcal{T}}_{\text{used}}(\varepsilon)$, which is computed using the optimal weight function $\widehat{v}_{\text{opt}}(\varepsilon, x)$. The terms $\widehat{F}_0(x)$, $\widehat{F}(x)$, and $\widehat{F}_{\boldsymbol{P}}(x)$ denote the CDFs of the null, observed, and watermarked samples, respectively.

Remark B.1. *The only difference between $\widehat{\mathcal{T}}_{\text{used}}$ and $\widehat{\mathcal{T}}$ (defined in (8)) is that the former approximates $\int \widehat{v}_{\text{opt}}(\varepsilon, x)\mathrm{d}F_0(x)$ using finite samples, while the latter employs exact integration, primarily for theoretical analysis.*

To solve this problem, we use the "scipy.optimize.minimize" function from the Python SciPy library [33]. The objective function $\mathcal{L}(\varepsilon)$ is specified in (39), with bounds for ε constrained to the interval $[\varepsilon_{\min}, 1 - \varepsilon_{\min}]$. We employ the default gradient-free optimization method, "L-BFGS-B", to ensure robust and efficient convergence. To identify a suitable initial point for the optimization, we iteratively compose the operator $\widehat{\mathcal{T}}_{\text{used}}$, starting from 0.9. Specifically, we compute $\widehat{\mathcal{T}}_{\text{used}} \circ \cdots \circ \widehat{\mathcal{T}}_{\text{used}}(0.9)$ with 20 compositions, using the resulting value as the starting point.

Algorithm 1 Generate NTP distributions

Input: Dominance parameter Δ, vocabulary size $|\mathcal{W}|$.
Sample parameters $a \sim \text{Uniform}(0.95, 1.5)$ and $b \sim \text{Uniform}(0.01, 0.1)$.
Sample the support size $k \sim \text{DiscreteUniform}[5, 15]$.
Generate a Zipf-like head vector: $\boldsymbol{h} = \text{Zipf}(a, b, k)$, where $\boldsymbol{h} \in \mathbb{R}^k$ is a probability vector.
Sample the random dominance parameter $\Delta_{\text{used}} \sim \text{Uniform}(0, \Delta)$.
Define the scaling factor $b = \frac{1-\Delta_{\text{used}}}{\max_{i\in[k]} h_i}$, where $\boldsymbol{h} = (h_1, \ldots, h_k)$.
Initialize $\boldsymbol{P} = \mathbf{0}_{|\mathcal{W}|}$.
if $b \leq 1$ **then**
Assign the first k entries of $\boldsymbol{P}$ to $b \cdot \boldsymbol{h}$, and distribute the remaining probability uniformly:

$$\boldsymbol{P}_{1:k} = b \cdot \boldsymbol{h}, \quad \boldsymbol{P}_{(k+1):|\mathcal{W}|} = (1-b) \cdot \frac{\mathbf{1}_{|\mathcal{W}|-k}}{|\mathcal{W}| - k}.$$

else
Assign the first entry of $\boldsymbol{P}$ to $1 - \Delta_{\text{used}}$, the next k entries to $\frac{\Delta_{\text{used}}}{2} \cdot \boldsymbol{h}$, and distribute the remaining probability uniformly:

$$P_1 = 1 - \Delta_{\text{used}}, \quad P_{2:(k+1)} = \frac{\Delta_{\text{used}}}{2} \cdot \boldsymbol{h}, \quad \boldsymbol{P}_{(k+2):|\mathcal{W}|} = \frac{\Delta_{\text{used}}}{2} \cdot \frac{\mathbf{1}_{|\mathcal{W}|-k-1}}{|\mathcal{W}| - k - 1}.$$

end if
Return: A randomly permuted version of $\boldsymbol{P}$.

Construction of NTP distributions. We adopt a similar approach to [19] for constructing NTP distributions, as detailed in Algorithm 1. The function $\text{Zipf}(a, b, k)$ generates a Zipf-like probability distribution over a specified support size k, where the probabilities are computed as $(i + b)^{-a}$ for $i = 1, \ldots, k$ and normalized to sum to 1. The NTP distributions generated by Algorithm 1 are parameterized by Δ, which controls the dominance of the largest probability, ensuring it remains below $1 - \Delta$ uniformly. The constructed distributions balance a dominant "head" component $\boldsymbol{h}$, generated using the Zipf distribution, and a uniform "tail" component, ensuring the entire distribution sums to 1. For all simulation experiments, we set the vocabulary size to $|\mathcal{W}| = 10^3$.

Failure on green-red list watermarks. As suggested by the original work [15], we set the parameters $(\gamma, \delta) = (0.5, 2)$ for the green-red list watermarks, which strike a balance between watermark signal strength and data distortion. As discussed in Sections 4.1 and 4.2, our estimators INI and IND are highly biased when NTP distributions $\boldsymbol{P}_{1:n}$ are inaccessible, likely producing estimates outside the feasible domain $[0, 1]$. To ensure meaningful comparison in Figure 1d, we plot the errors of projected estimators $[\widehat{\varepsilon}_{\text{ini}}(\delta)]_{[0,1]}$ and $[\widehat{\varepsilon}_{\text{rfn}}]_{[0,1]}$, which guarantees that MAEs do not exceed 1. One can easily show that the accuracy analysis for MAEs in Theorem 4.1 and 4.2 still applies due to the nonexpansiveness of the projection operator.

B.2 Empirical CDFs of Mixture Distributions

Figure 3 illustrates the CDFs of the mixture distribution $(1 - \varepsilon)F_0 + \varepsilon\bar{F}_{\boldsymbol{P}}$ for the Gumbel-max watermark, while Figure 4 shows the corresponding CDFs for the inverse transform watermark.

The alternative CDF, corresponding to $\varepsilon = 1$, is consistently stochastically larger than the null distribution, which corresponds to the case where $\varepsilon = 0$. As ε increases, the mixture CDF becomes progressively more right-skewed.

In constructing the NTP distributions, we ensure that each $\boldsymbol{P}_t$ is Δ-regular, meaning the largest probability in $\boldsymbol{P}_t$ is strictly less than $1 - \Delta$. Specifically, this largest probability is uniformly distributed within $(0, 1 - \Delta)$. Consequently, a larger value of Δ induces a greater discrepancy between F_0 and $\bar{F}_{\boldsymbol{P}}$, leading to increased right-skewness in the mixture CDF.

B.3 Additional Results on Estimation Accuracy

Figures 5 and 6 show the complete results of the estimation performance of various estimators on the Gumbel-max and inverse transform watermarks, respectively. In most cases, OPT achieves the lowest MAE.

B.4 Additional Results on Inherent Bias

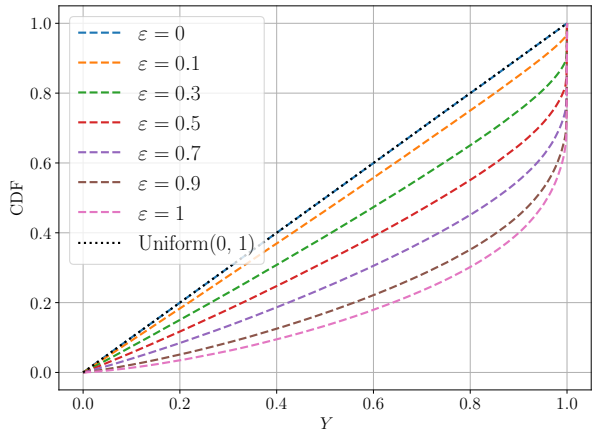
We vary the values of (γ, μ) and plot the estimators obtained by solving the L_2-regularized MLEs described in Theorem 4.5 in Figure 7. In each subplot, we fix the L_2 regularization coefficient to $\lambda = 10^{-2}$ and consider three sample sizes: $n \in \{10^3, 10^4, 10^5\}$. The red curves represent the theoretical limit proportion $\widehat{\varepsilon}$ from Theorem 4.5, while the other colored curves depict the MLE estimates $\widehat{\varepsilon}_\lambda$ for the different sample sizes. Across all combinations of (γ, μ), the L_2-regularized MLEs align closely with the theoretical limit proportion. This alignment validates our theory and provides strong support for the existence of inherent bias.

C Open-source LLM Experiments: Details and Results

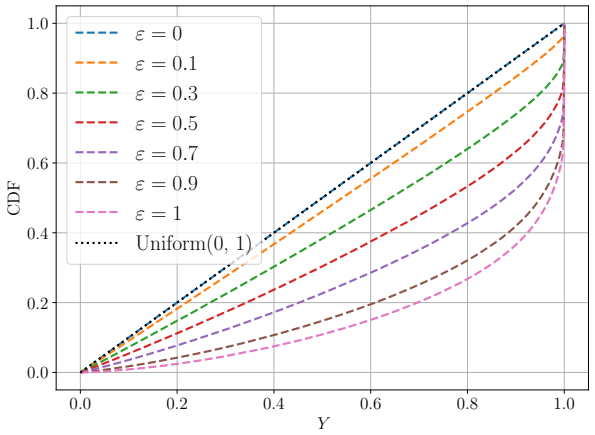
C.1 Details on Mixture Data Setting

Watermark generation. We use a context window of size $m = 4$, where the random variable ζ_t is derived as $\zeta_t = \mathcal{A}(s_{(t-m):(t-1)}, \mathtt{Key})$, based on the preceding m tokens. The randomness is generated using the hash function $\mathcal{A}$, as introduced in [43]. To avoid repetitive outputs, a watermark is applied only when the current text window is unique within the generation history, following the strategies outlined in [14, 36, 8]. This technique is known as 1-sequence repeated context masking in [8]. If no watermark is applied, tokens are sampled directly from the NTP distribution using multinomial sampling. When constructing a dataset with the true proportion ε, we aim for approximately ε-proportion of the samples (i.e., observed pivotal statistics) to be watermarked and $(1 - \varepsilon)$-proportion to be null samples. So we independently watermark each token with probability ε. However, applying 1-sequence repeated context masking can reduce the effective watermark proportion, as it may prevent some tokens that should be watermarked from being marked as such. As a result, in our experiments, we calculate the average number of watermarked tokens to determine the ground truth ε.

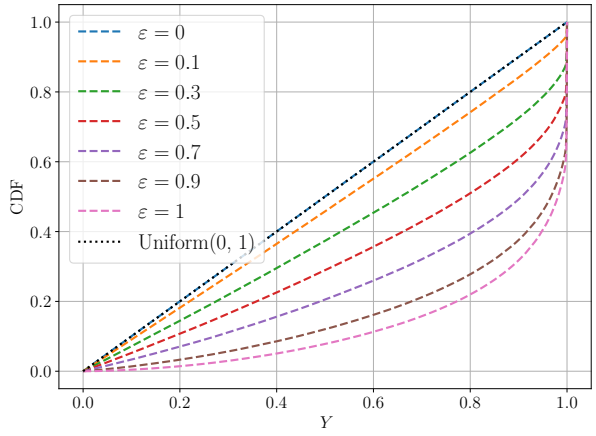
Prompt generation. Our prompt generation approach largely follows the methodology described in Appendix C.1 of [19]. Prompts are created by sampling documents from the news-focused section of the C4 dataset [27]. Each document is required to have at least 50 tokens, and shorter documents are excluded. Since retokenization may not perfectly align with the original token counts, we ensure



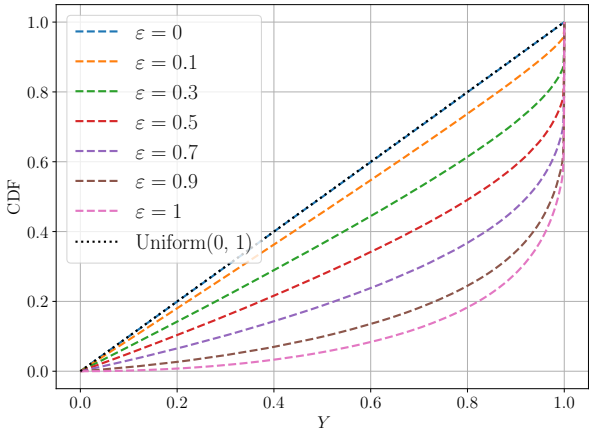
(a) $\Delta = 0.1$



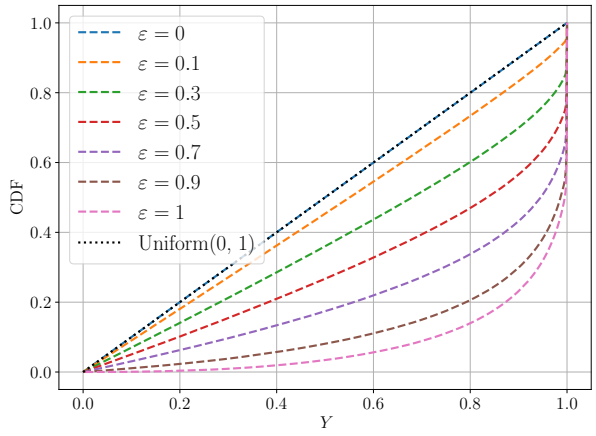
(b) $\Delta = 0.2$



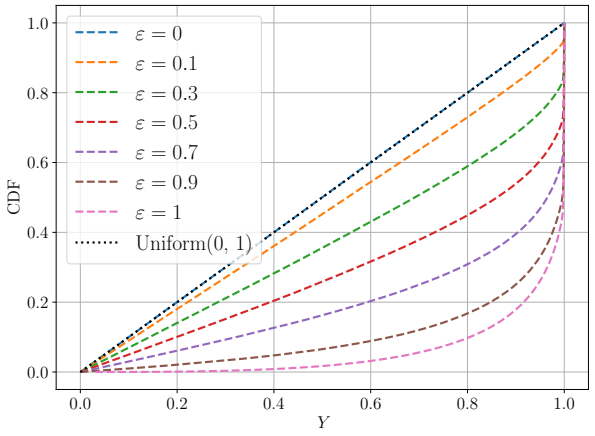
(c) $\Delta = 0.3$



(d) $\Delta = 0.4$

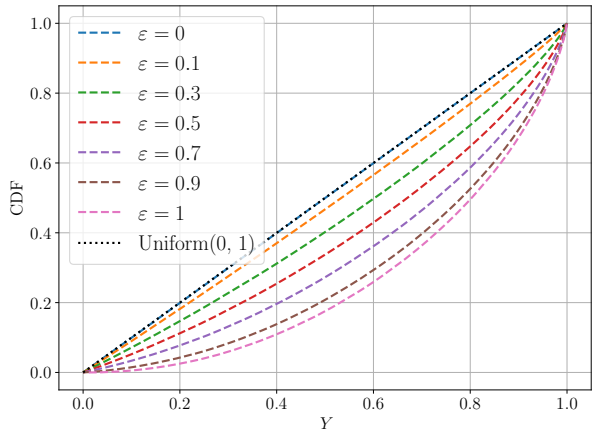


(e) $\Delta = 0.5$

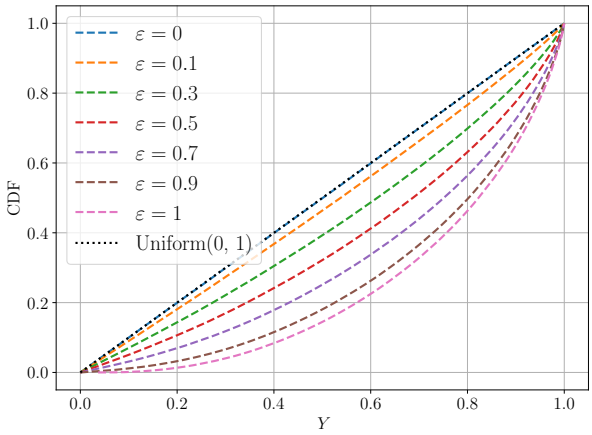


(f) $\Delta = 0.6$

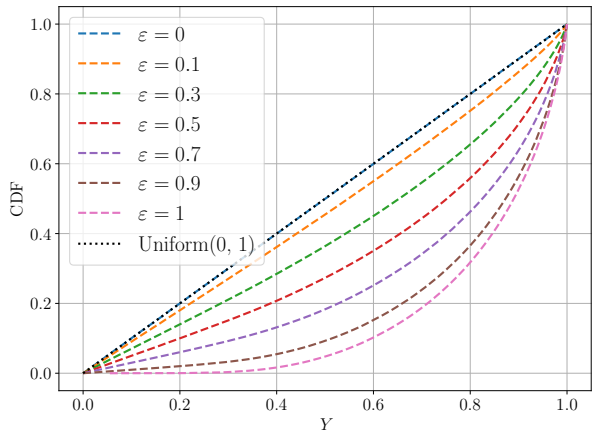
Figure 3: CDF plots of mixture distributions for the Gumbel-max watermark. Here, Δ represents the dominance parameter used in Algorithm 1 for generating NTP distributions, and ε denotes the true proportion.



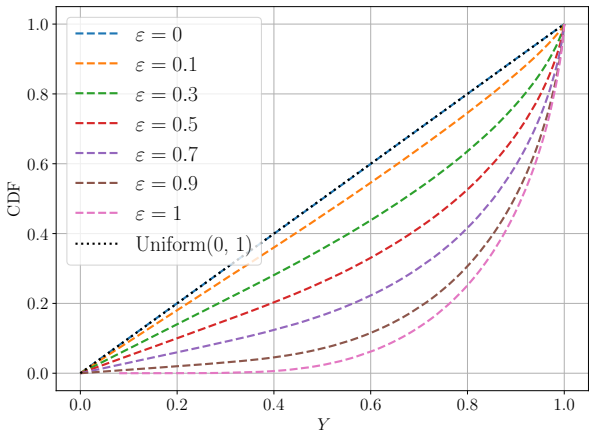
(a) $\Delta = 0.1$



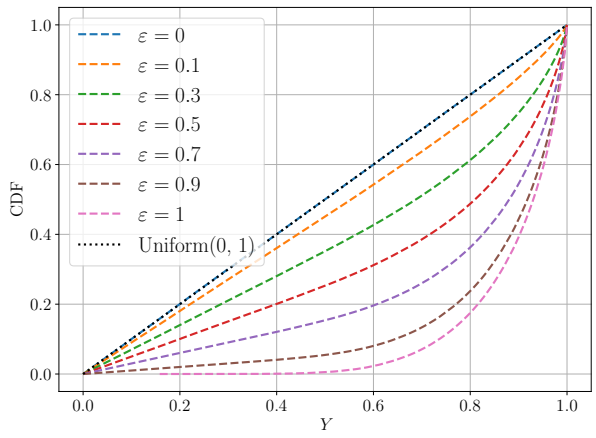
(b) $\Delta = 0.2$



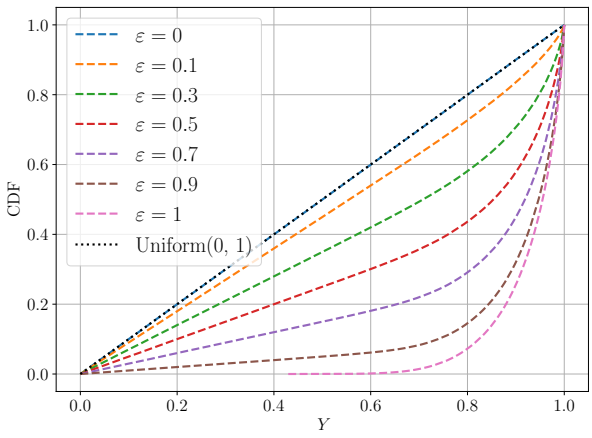
(c) $\Delta = 0.5$



(d) $\Delta = 0.6$

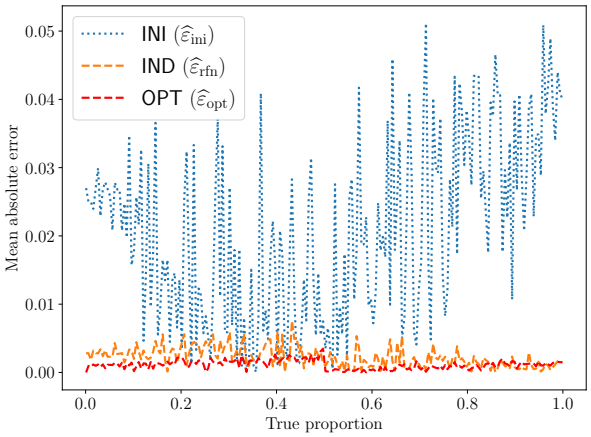


(e) $\Delta = 0.7$

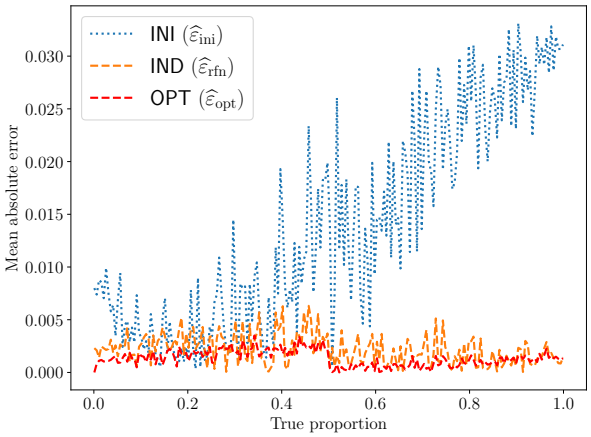


(f) $\Delta = 0.8$

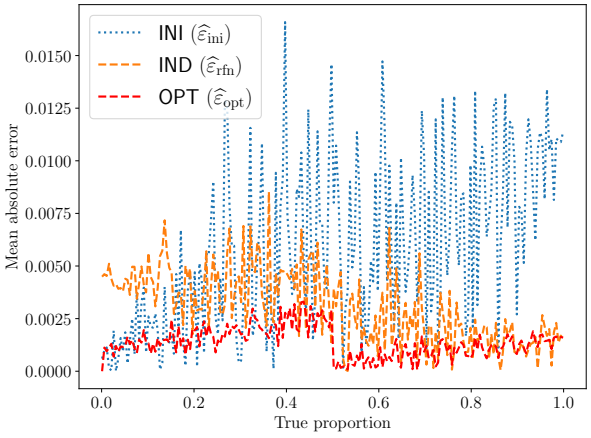
Figure 4: CDF plots of mixture distributions for the inverse transform watermark. Here, Δ represents the dominance parameter used in Algorithm 1 for generating NTP distributions, and ε denotes the true proportion.



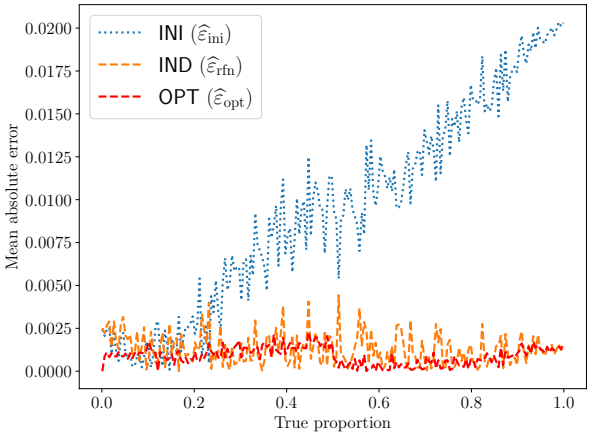
(a) $\Delta = 0.1$



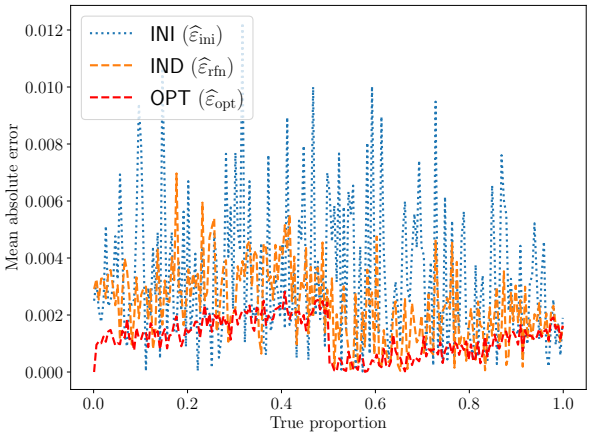
(b) $\Delta = 0.2$



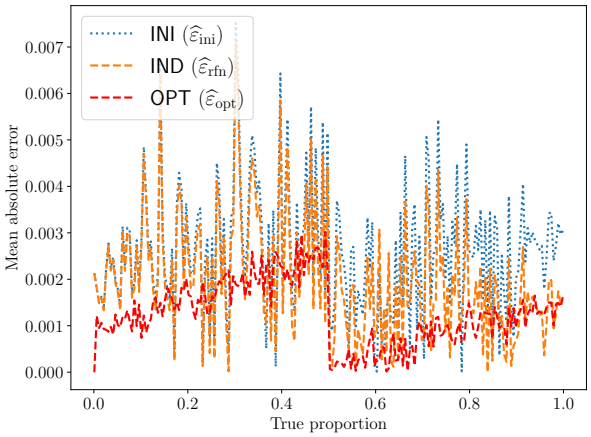
(c) $\Delta = 0.3$



(d) $\Delta = 0.4$

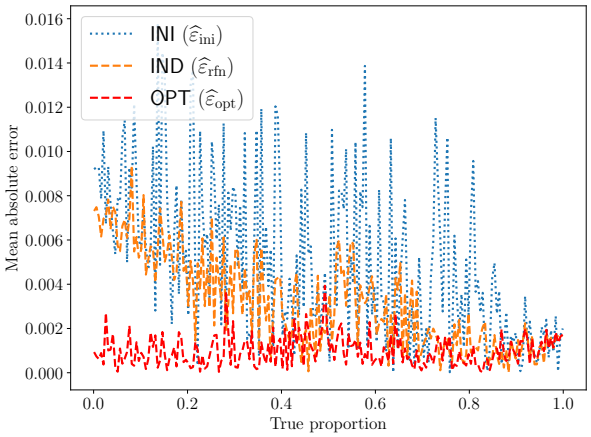


(e) $\Delta = 0.5$

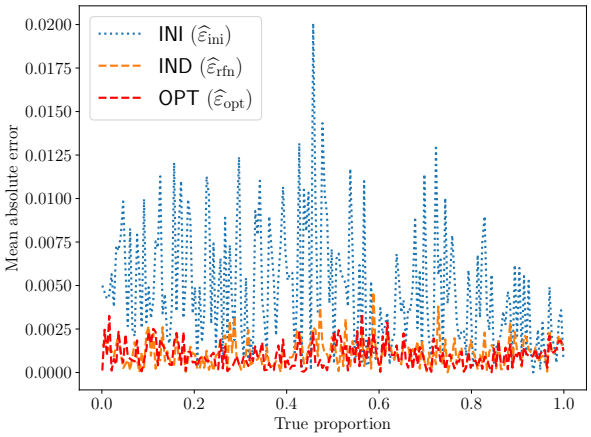


(f) $\Delta = 0.6$

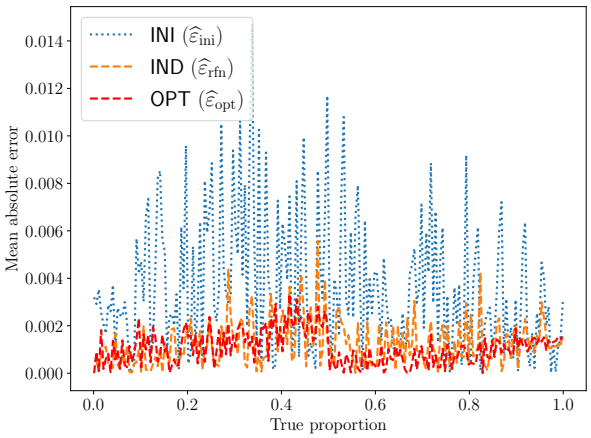
Figure 5: Mean absolute errors (MAEs) for the Gumbel-max watermark. Here Δ is the dominance parameter in Algorithm 1 for the NTP distributions. The best MAEs for both INI $\widehat{\varepsilon}_{\text{ini}}(\delta)$ and IND $\widehat{\varepsilon}_{\text{rfn}}(\delta)$, are reported by searching δ over $\{10^{-1}, 10^{-2}, 10^{-3}\}$.



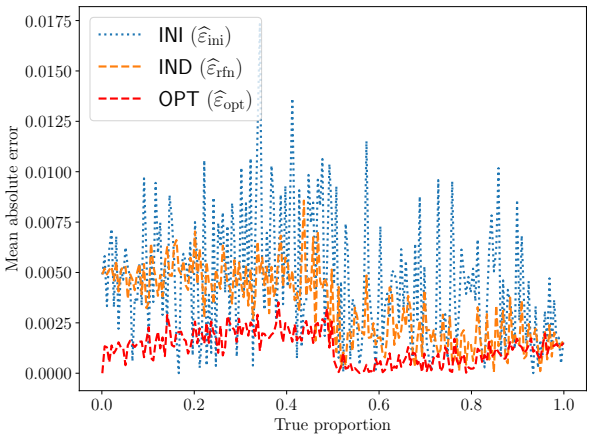
(a) $\Delta = 0.1$



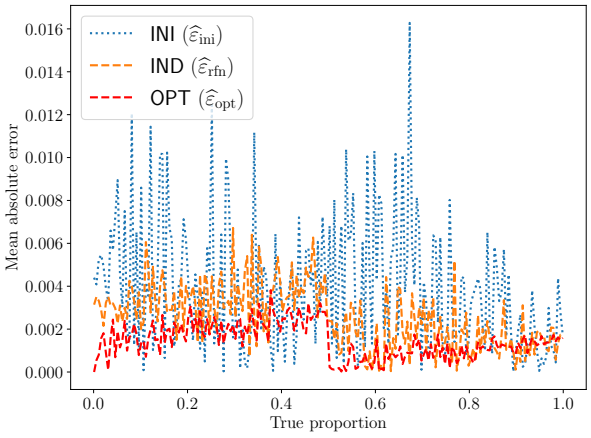
(b) $\Delta = 0.2$



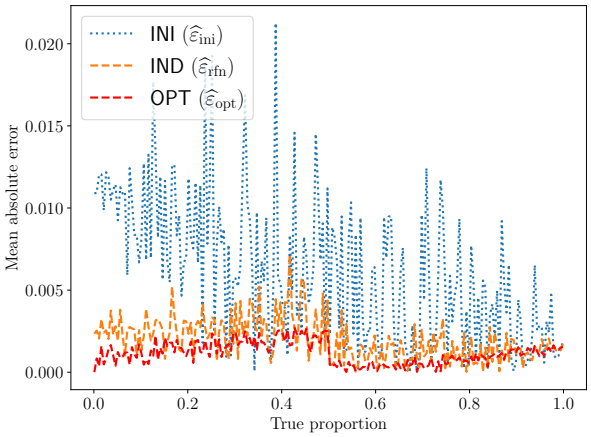
(c) $\Delta = 0.5$



(d) $\Delta = 0.6$

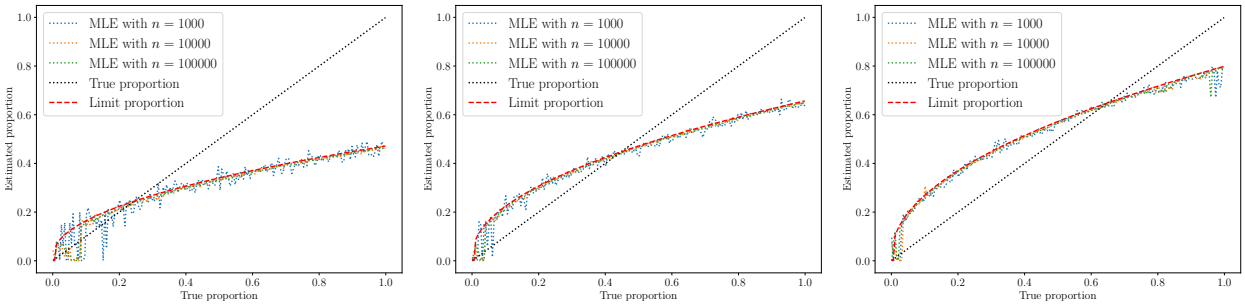


(e) $\Delta = 0.7$

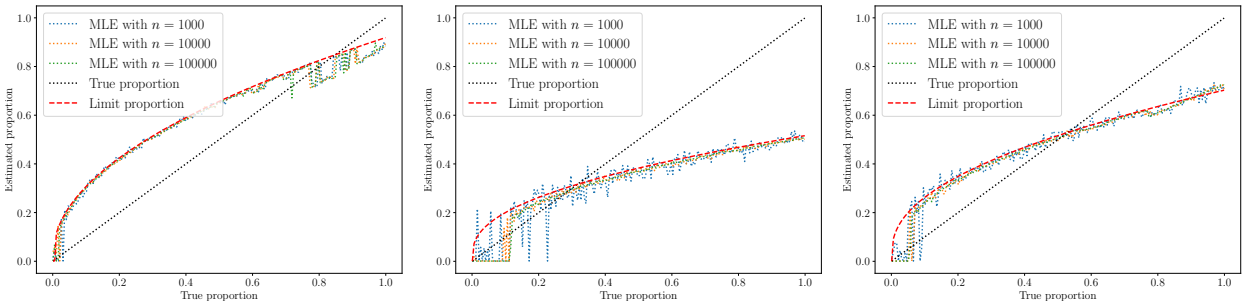


(f) $\Delta = 0.8$

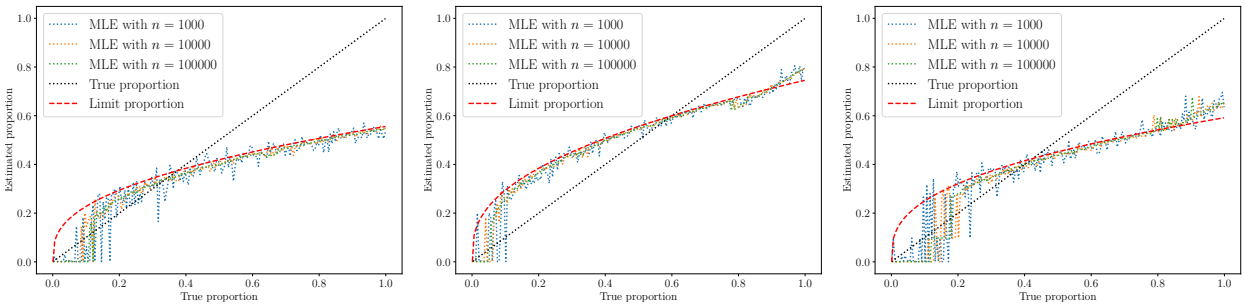
Figure 6: Mean absolute errors (MAEs) for the inverse transform watermark. Here Δ is the dominance parameter in Algorithm 1 for the NTP distributions. The best MAEs for both INI $\widehat{\varepsilon}_{\text{ini}}(\delta)$ and IND $\widehat{\varepsilon}_{\text{rfn}}(\delta)$, are reported by searching δ over $\{10^{-1}, 10^{-2}, 10^{-3}\}$.



(a) $(\gamma, \mu) = (0.1, 0.3)$ (b) $(\gamma, \mu) = (0.1, 0.5)$ (c) $(\gamma, \mu) = (0.1, 0.7)$



(d) $(\gamma, \mu) = (0.1, 0.9)$ (e) $(\gamma, \mu) = (0.3, 0.5)$ (f) $(\gamma, \mu) = (0.3, 0.7)$



(g) $(\gamma, \mu) = (0.5, 0.7)$ (h) $(\gamma, \mu) = (0.5, 0.9)$ (i) $(\gamma, \mu) = (0.7, 0.9)$

Figure 7: More results illustrating the inherent bias caused by non-identifiability. We vary the values of (γ, μ). The red curves represent the limit proportion from Theorem 4.5, while the other colored curves show the MLE estimates for different sample sizes n.

that the verifier always processes at least n tokens by adding special padding tokens, which are model-dependent. Additionally, to reduce the need for padding, we generate extra tokens—referred to as buffer tokens—beyond n. In all experiments, 20 buffer tokens are generated, which generally eliminates the necessity for padding.

Details for IND and OPT. The computation for the refined estimators (namely IND and OPT) follows the procedure outlined in Appendix B. To evaluate MAEs across various proportions ε, we generate a mixed-source dataset for each trial, ensuring that approximately ε-proportion of the samples are watermarked. Importantly, these datasets consist solely of pivotal statistics, not tokens. After generating datasets for different values of ε, we aggregate all the watermarked pivotal statistics into a fully watermarked dataset. This combined dataset is then used to estimate $\widehat{g}$ and $\widehat{F}_{\boldsymbol{P}}$. As a result, some pivotal statistics are entirely relevant to the current estimation task.

Details for the baseline WPL. WPL, proposed by Zhao et al. [44], uses an existing online learning algorithm to compute a watermark score for each token, classifying it as watermarked if the score exceeds a specified threshold. For the Gumbel-max watermark, the score is defined as $-\log(1-Y)$, while for the inverse transform watermark, it is simply Y. The corresponding pivotal statistics Y are summarized in Table 1. To ensure a fair and competitive comparison, we tune the threshold over a predefined set and report the best result that achieves the lowest MAE across both figures and tables. Specifically, the threshold set is $\{1.3, 1.5, 1.7\}$ for Gumbel-max and $\{0.73, 0.75, 0.77\}$ for the inverse transform watermark.

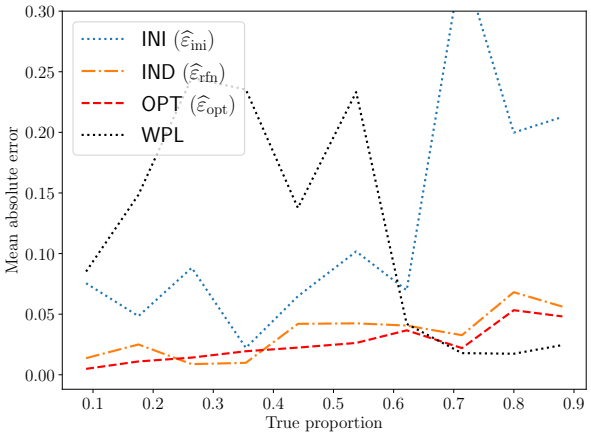
C.2 Additional Results on Estimation Accuracy

Figures 8 and 9 illustrate the estimation performance of various estimators on the OPT-13B and LLaMA3.1-8B models using the C4 dataset, respectively. Figure 10 presents the performance on the arXiv dataset. In most scenarios, OPT achieves the lowest MAE, while WPL exhibits poor accuracy when the true proportion is around 0.5.

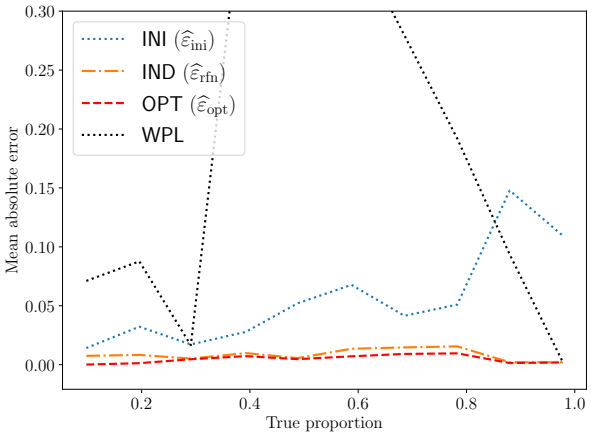
C.3 Details on Post-processing Modification

The post-processing modification setup closely follows the earlier mixture data setting. We use a context window of size $m = 4$ and fix the temperature to 1. The same three LLMs are evaluated on both the C4 and arXiv datasets. Since the high temperature introduces sufficient randomness, we didn't apply 1-sequence repeated context masking [8]. For each run, we sample prompts using the last 50 tokens of randomly selected documents and generate 500 continuation tokens. A specified fraction of tokens is then modified via random substitution, deletion, or insertion, and we apply various methods to estimate the watermark proportion.

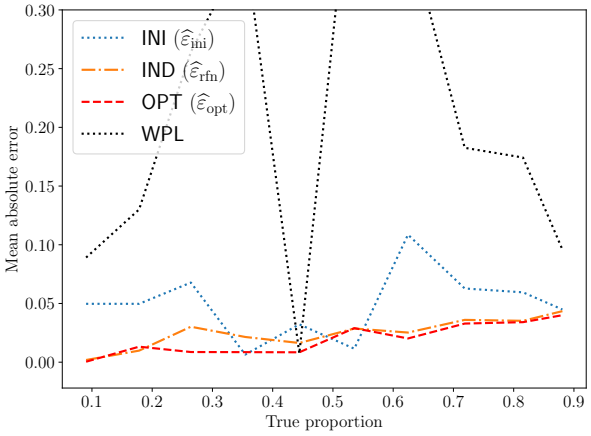
The ground-truth proportion is defined as the fraction of pivotal statistics following the alternative distribution. Specifically, for any segment $w_{(t-m):t}$ that appears in the original modified text, the corresponding statistic $Y_t = Y(w_t, \zeta_t) = Y(w_t, \mathcal{A}(w_{(t-m):(t-1)}, \texttt{Key}))$ is treated as watermarked. This is equivalent to assuming $Y_t | \boldsymbol{P}_t \sim \mu_{1,\boldsymbol{P}_t}$. We compute the proportion of such pivotal statistics as the true ε.



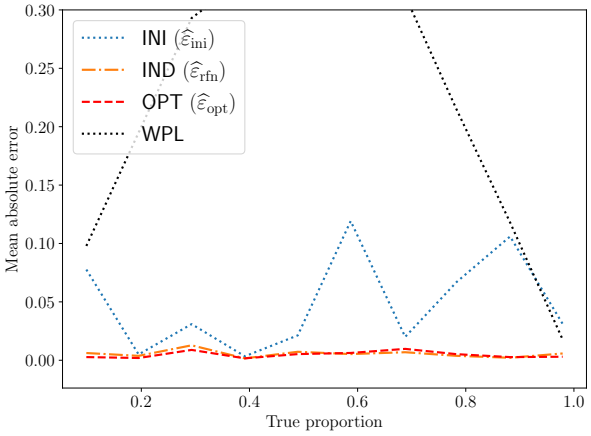
(a) Gumbel-max with T = 0.7



(b) Gumbel-max with T = 1

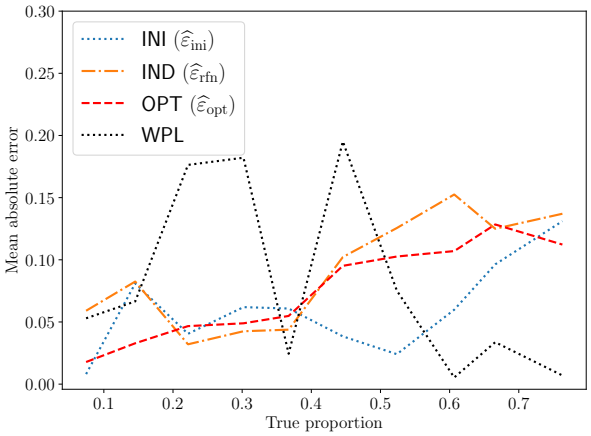


(c) Inverse with T = 0.7

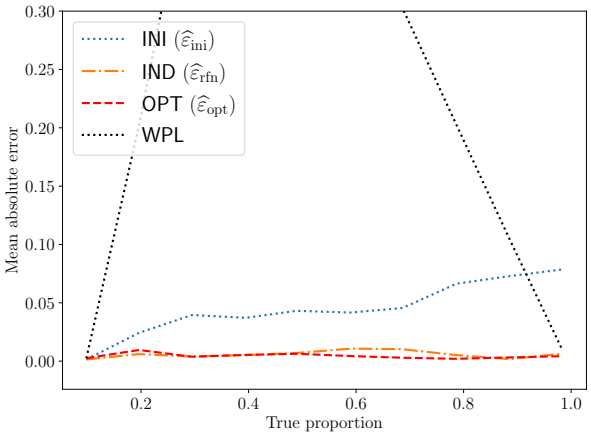


(d) Inverse with T = 1

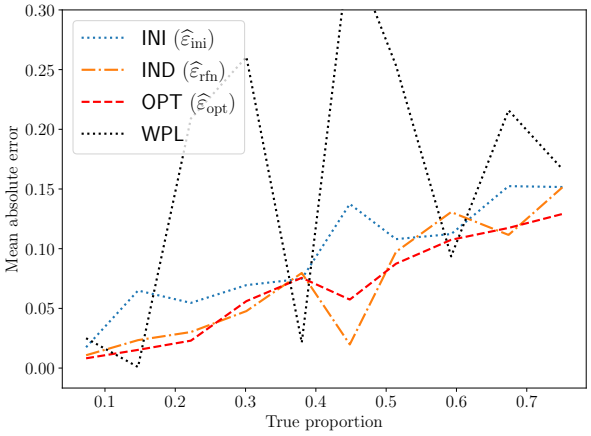
Figure 8: Mean absolute errors of estimators on the C4 dataset, evaluated across various true proportions ε and temperature parameters for the OPT-13B model.



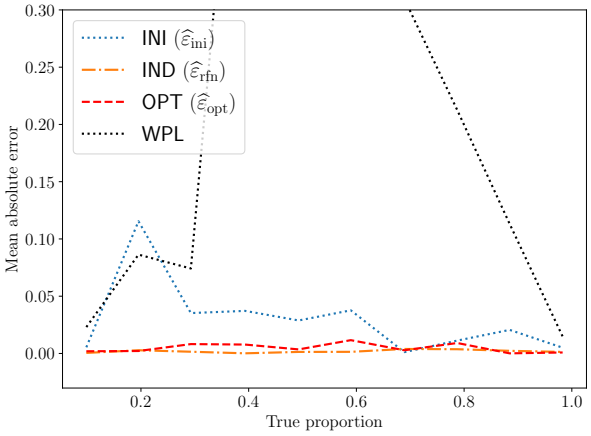
(a) Gumbel-max with T = 0.7



(b) Gumbel-max with T = 1

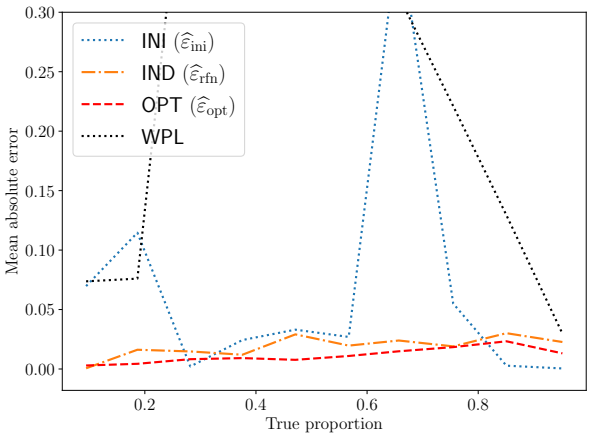


(c) Inverse with T = 0.7

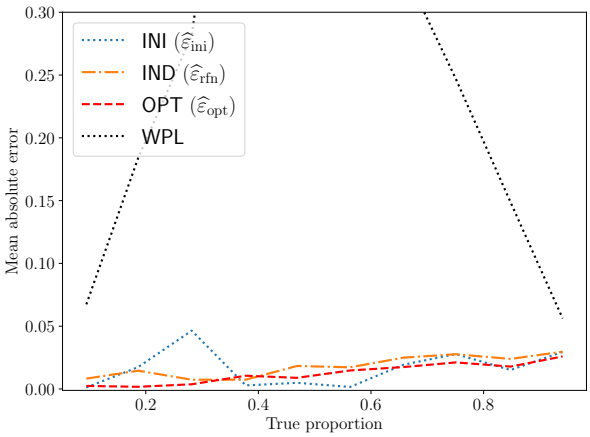


(d) Inverse with T = 1

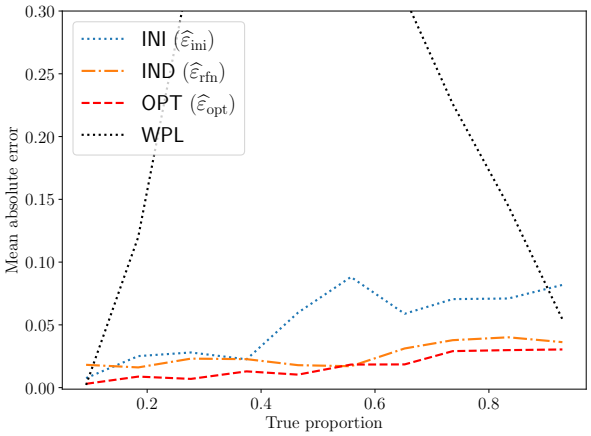
Figure 9: Mean absolute errors of estimators on the C4 dataset, evaluated across various true proportions ε and temperature parameters for the LLaMA3.1-8B model.



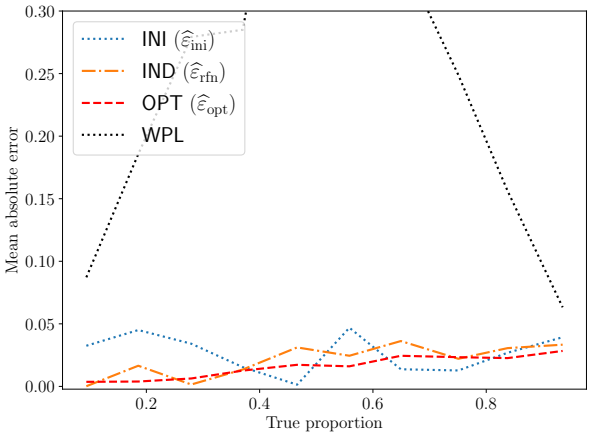
(a) Gumbel-max on OPT-1.3B



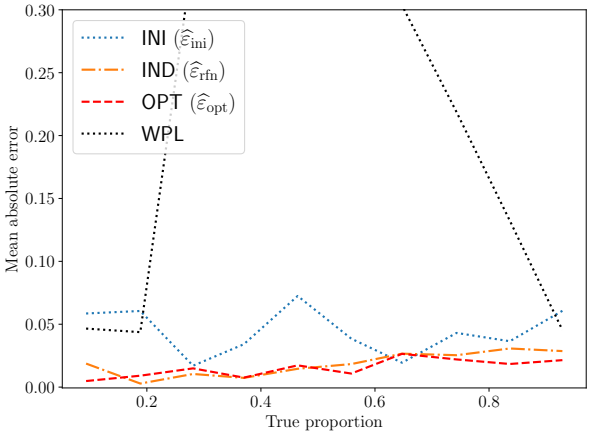
(b) Inverse on OPT-1.3B



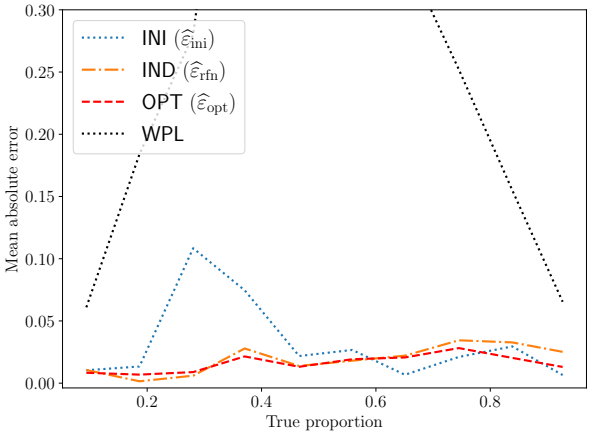
(c) Gumbel-max on OPT-13B



(d) Inverse on OPT-13B



(e) Gumbel-max on LLaMA3.1-8B



(f) Inverse on LLaMA3.1-8B

Figure 10: Mean absolute errors of estimators on the arXiv dataset, evaluated across various true proportions ε for the three different models. The temperature parameter is set to 1.

C.4 Additional Results on Post-processing Modification

Table 5 reports the average MAEs under the three post-processing modifications on the arXiv dataset, serving as the counterpart to Table 4. Once again, our estimator OPT achieves the lowest MAE and variance in most cases, though INI or IND may occasionally perform better.

Table 5: Averaged MAEs under common modification calculated over 11 ground truth ε values on open-sources model experiments. Standard deviations are provided in parentheses, and all values are reported in units of 10^{-3}. Bold numbers denote the best performance.

Models	Edit types	Gumbel-max				Inverse transform			
		WPL	INI	IND	OPT	WPL	INI	IND	OPT
OPT-1.3B	Substitution	125(82)	29(28)	5(3)	**1(1)**	266(113)	37(22)	4(3)	**2(1)**
	Insertion	139(80)	41(39)	**7(4)**	9(5)	263(104)	34(22)	**9**(7)	10**(5)**
	Deletion	224(91)	**38**(41)	74(20)	73**(19)**	280(101)	**32**(22)	72(19)	74**(18)**
OPT-13B	Substitution	90(53)	77(34)	2(2)	**1(1)**	251(101)	56(34)	5(3)	**1(1)**
	Insertion	95(58)	53(42)	12**(5)**	**10(5)**	243(82)	46(29)	10(7)	**9(5)**
	Deletion	160(82)	**44**(41)	75(19)	73**(17)**	247(102)	**61**(51)	75(18)	77**(19)**
LLaMA-8B	Substitution	63(42)	79(41)	3(3)	**1(1)**	239(78)	58(38)	5(3)	**1(1)**
	Insertion	67(43)	83(58)	9(7)	**3(3)**	227(85)	47(31)	7(6)	**3(2)**
	Deletion	125(63)	**42**(35)	61**(21)**	48(23)	239(86)	**47**(40)	62**(20)**	**47**(23)